

NS Branch Book
Navier–Stokes Regularity

Elements of Nested Fibrational Cosmology —
Derived Branch

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REMARK 0.1 ([R]). Current Status Capsule. *Current posture:* CERT-PROJ with the obstruction/ledger common-state frontier open (MIG-052): the NS.6.1/NS.7.1 conditional chain is complete on the ledger side, and its transfer to the window obstruction — hence the endpoint — passes through the open common-state bridge obligation (Obl. 9.5); the former domain-bounded conditional CERT-CLOSE posture (MIG-050 wording) is superseded and may not be cited; unconditional global regularity (the Clay Stage-3 question) remains the external frontier.

Authoritative source: 15.1 (final status proposition (current posture)).

*Frontier accountings (never summed): syntactic [O] in this branch — 1 (Obl. 9.5); curated named-obligation roster — ob:NS-common-state-bridge, with IDC (1)–(2) recorded under it as subsidiary and insufficient and α (**hyp:NS-alpha**) withdrawn; reduced irreducible frontier — the internal common-state bridge and the external unconditional-regularity question; branch posture — CERT-PROJ, open common-state frontier.*

Constitutional intake: Front-block material below is constitutional intake, not current status.

Front Block

The following five items constitute the mandatory intake declaration required by Book VII, Theorem VII.8.4 (Canonical Closure Schema Theorem) and Book V, Corollary V.6.2.

Imported spine results.

- (1) Book III: Observational Transfer Theorem of the Spine; Unified Coercive-Transport Inequality (Theorem III.6.1).
- (2) Book IV: Universal Source Theorem (Theorem IV.7.1).
- (3) Book V: Universal Branch Legitimacy Theorem (Theorem V.6.1); Branch Projection Theorem (Proposition V.3.1).
- (4) Book VI: Continuum Bridge Schema Theorem (Theorem VI.6.4); Effective Continuum Legitimacy Theorem (Theorem VI.8.1); Continuum No-Smuggling Corollary (Theorem VI.7.2).
- (5) Book VII: Scoped Certificate Rule (Theorem VII.2.4); Promotion Requires Transfer Theorem (Theorem VII.2.5); Named Failure Frontier Theorem (Theorem VII.4.3); Canonical Closure Schema Theorem (Theorem VII.6.4).

Branch-specific data.

- (1) A declared NS target regime: the continuum domain of the Navier–Stokes equation whose regularity is the endpoint target.
- (2) A branch-specific NS observable family, defined in Section 2.
- (3) A branch-specific NS obstruction quantity, defined in Section 4.
- (4) A branch-specific NS continuation criterion, defined in Section 5.
- (5) An active transport-weighted averaging operator and window structure, defined in Section 4.

Endpoint target. The endpoint target is the lawful NS descendant closure claim: a continuum Navier–Stokes regularity result (smoothness of the Leray weak solution

$u \in L_t^\infty H_x^1$) in the declared NS target regime, if the required bridge stack is fully discharged.

Bridge stack. The NS branch requires the following bridge steps in sequence:

- (1) source-descended transported coercive structure;
- (2) NS obstruction control;
- (3) NS continuation criterion;
- (4) NS regularity endpoint.

The first step is supplied by Book III. The second and third steps require the open bridge theorem NS.6.1. The fourth step is the bridge theorem NS.7.1.

Current status. **Frontier-blocked lawful branch.**

The NS branch is a lawful descendant of the canonical spine and a mathematically meaningful branch program. It is currently blocked from closure by two named frontier items:

- (1) *Proof-program frontier*: Theorem NS.Ren.5 (Late Alignment-to-Dissipation Lemma) — the smallest unresolved theorem still plausibly inside the current active architecture.
- (2) *Structural-input frontier*: NS.SB2 (Positive Reserve on the Safe Tail) — a structural condition on late-tail boundary behavior that may require a genuinely new ingredient beyond the current architecture.

NS.0 Purpose

This branch studies a lawful descendant of the universal source object $\mathbf{U}$ under the branch-legitimacy law of Book V. Its task is to determine whether the declared NS endpoint target — continuum Navier–Stokes regularity — can be realized by a certified source-descended projection together with the explicit NS bridge stack stated in the front block above.

This branch is especially important because it tests the full canonical discipline simultaneously: lawful descent from $\mathbf{U}$, application of the Unified Coercive-Transport Inequality from Book III, honest use of the Book VI continuum interface machinery, and named frontier discipline from Book VII. A program that handles all four correctly is genuinely rigorous; any program that blurs any of them is not.

This branch book does not function as an alternate foundation for NFC. It claims only the descendant force explicitly licensed by its declared spine imports, bridge stack, and stated branch status.

1. Imported Spine Structure

STANDING RULE 1.1 ([D]). The NS branch does not function as an alternate foundation. All source-level structure enters from the canonical spine through the declared imports; no source-level claim originates within this branch.

STANDING RULE 1.2 ([D]). No NS branch claim may be promoted beyond its declared scope without satisfying the Continuum Bridge Schema Theorem (Book VI,

Theorem VI.6.4) and the Promotion Requires Transfer Theorem (Book VII, Theorem VII.2.5).

The branch imports:

- from Book III: source-level transported coercivity, in the form of the Unified Coercive-Transport Inequality $C_{n+1} \leq \Theta_n C_n + E_n + B_n$;
- from Book IV: lawful source descent from $\mathbf{U}$;
- from Book V: branch admissibility and legitimacy certification;
- from Book VI: the continuum-interface law governing when any smooth language used below is licensed; and
- from Book VII: status, frontier, and promotion discipline.

No further upstream authority is presumed.

2. Branch-Specific Arena

DEFINITION 2.1 ([D]). An *NS branch candidate* is a lawful descendant branch whose endpoint target is a declared continuum Navier–Stokes regularity regime: a connected open domain $\Omega \subseteq \mathbb{R}^3$ together with a Leray weak solution u on a declared time interval, and an explicit regularity criterion (smoothness of u in the sense $u \in L_t^\infty H_x^1$ on that interval).

DEFINITION 2.2 ([D]). The *NS observable family* is the branch-specific descendant observable family on which the NS bridge stack acts. It is a certified observable family in the sense of Book VI, Definition VI.2.1, obtained by restricting the source-level persistent observable structure of Book III to the NS branch candidate.

DEFINITION 2.3 ([D]). The *NS endpoint target* is the declared continuum regularity criterion: the assertion that the Leray weak solution u belongs to $L_t^\infty H_x^1$ on the declared time interval.

DEFINITION 2.4 ([D]). The *declared NS defect burden* is the branch-specific burden recording the part of the descendant regime whose continued accumulation obstructs lawful continuation toward the NS endpoint target. It is defined in terms of the NS observable family and the boundary-defect structure inherited from Book III.

REMARK 2.5 ([R]). The NS branch must keep three levels distinct throughout:

- (1) source-descended obstruction control,
- (2) promoted continuation structure, and
- (3) final endpoint closure.

These levels correspond respectively to: what Book III supplies, what NS.6.1 must provide, and what NS.7.1 must provide. No step may be collapsed into the previous one without a separate proof.

DEFINITION 2.6 ([D]). (Strong Boundary Approximation.) The NS branch satisfies *Strong BA* if

$$N_{\text{cert}}(U_n, t) = O(|\partial U_n|)$$

uniformly in n and t : the count of certified defect events in the Van Hove exhaustion window U_n is at most proportional to the boundary size $|\partial U_n|$, not to the volume

$|U_n|$. Strong BA is the boundary-order condition; it is equivalent to NS Stage-3 closure (global regularity of Leray weak solutions) via the obstruction-decay chain

$$\text{Strong BA} \iff \sup_n Q_n < \infty \iff D_{\text{floor}}(U_n) \leq C_* a_n^2 \iff \text{NS Stage-3 closure.}$$

DEFINITION 2.7 ([D]). (NS Master Inequality.) The *NS master inequality* is

$$\nu \cdot c_* > 2 \cdot C_{\text{NL}}(s) \cdot \|u_0\|_{H^s},$$

where $\nu > 0$ is the viscosity parameter (table-computable via KPO-4: $\nu = \varphi_\infty / \xi C_2^2$), $c_* > 0$ is the coercivity constant from the Unified Coercive-Transport Inequality (Book III, Theorem III.5.1), and $C_{\text{NL}}(s) > 0$ is the NS nonlinear gate constant (finite, computable from the NF_2 type table). Under this inequality together with NS-A through NS-E: blowup is forbidden and global H^s regularity of the identified encoded NS limit holds. All quantities are finite predicates on NF_2 table data; no metric geometry is imported.

DEFINITION 2.8 ([D]). (NS-E: Stage-3 Identification Gate.) Let Φ be the *Stage-3 identification map*: the canonical map from the NS witness dynamics (the collar-level certified event process) to the classical NS energy-estimate framework. The obligation *NS-E* requires: Φ exists and is well-defined on the declared NS regime, so that the witness dynamics and the classical NS H^s energy estimate can be compared. NS-E is structurally prior to NS-C (surjectivity of Φ); it gates the bare existence before surjectivity is demanded.

LEMMA 2.9 ([C]). (NS-E Conditionally Discharged.) [Conditional on Phase-A, $K_0 = 7$, NS-4B Sobolev surrogate (Book VI, Thm. ??).] *The Stage-3 identification map Φ exists: the NS-4B Sobolev surrogate $\|f\|_\infty \leq C_S(s)\|f\|_{H^s}$ provides the canonical bridge from the discrete NF_2 witness dynamics to the classical H^s energy estimate framework, defining Φ as the composition of the collar-type encoding (KPO-3) with the Sobolev-type norm comparison.*

PROOF. By Book VI Thm. ??: the Sobolev surrogate $\|f\|_\infty \leq C_S(s)\|f\|_{H^s}$ holds with table-computable constants. The identification map Φ is defined by: $\Phi(u_{\text{discrete}}) = u_{\text{cont}}$, where u_{disc} is the discrete collar observable and u_{cont} is its Weyl-encoded continuum representative (via KPO-3). The Sobolev surrogate certifies that the H^s -norm of u_{cont} is bounded by a table-computable function of the discrete data. Hence Φ is well-defined on the declared NS regime. $\square$ $\square$

LEMMA 2.10 ([C]). (BUDGET Lemma as Derived Consequence.) [Conditional on Stage-0 thermodynamic density limit $f_\infty > 0$ and NS-A (scaling-critical coercion).] *The per-site billing bound $\sum_x \langle u_{k,x}, D_k u_{k,x} \rangle \leq B_k |S_k(\sigma)|$ is a theorem, not an independent hypothesis.*

PROOF. By Stage-0 (thermodynamic density): $f_\infty = \lim_n N_{\text{cert}}(U_n)/|U_n| > 0$ bounds the average per-site certified-event density. By NS-A (scaling-critical coercion): the coercive functional D_k contracts the per-site contribution at rate $\geq c_*$. Combining: $\sum_x \langle u_{k,x}, D_k u_{k,x} \rangle \leq c_*^{-1} f_\infty \cdot |S_k(\sigma)| \cdot \langle u, u \rangle / |U_k|$. Setting $B_k = c_*^{-1} f_\infty$ (table-computable) gives the stated bound. $\square$ $\square$

3. Lawful Descent and Endpoint Visibility

PROPOSITION 3.1 ([U]). *The NS branch is a lawful branch only if its endpoint target arises as a certified image of $\mathbf{U}$ under a declared realization structure.*

PROOF. By the Branch Projection Theorem of Book V (Proposition V.3.1), any lawful descendant endpoint must arise by certified descent from $\mathbf{U}$ through a declared realization functor. The NS branch must declare its realization structure and verify that the NS endpoint target is the image of $\mathbf{U}$ under that functor. $\square$ $\square$

PROPOSITION 3.2 ([U]). *The NS endpoint target has canonical force only if it admits a legitimacy witness.*

PROOF. By the Legitimacy Witness Requirement of Book V (Proposition V.3.2), endpoint force is canonical only when the endpoint datum is certified as depending solely on declared source descent, target structure, and declared branch-specific hypotheses. A legitimacy witness is required to certify this. $\square$ $\square$

COROLLARY 3.3 ([U]). *Any NS endpoint claim depending on undeclared analytic or interpretive supplementation is non-canonical.*

REMARK 3.4 ([R]). Propositions 3.1 and 3.2 establish that the NS branch is lawful only as a source-descended program, not as an independent continuum-first program. The branch may not quietly revert to treating the Navier–Stokes equation as a primitive object; every claim must trace back through the declared spine imports.

4. Source-Descended Obstruction Quantity

PROPOSITION 4.1 ([U]). *The NS branch inherits a source-descended transported coercive structure from Book III.*

PROOF. The Unified Coercive-Transport Inequality of Book III (Theorem III.6.1) is branch-agnostic and available to every lawful descendant. Since the NS branch is lawful by Proposition 3.1, it inherits that source-descended transported coercive structure. $\square$ $\square$

DEFINITION 4.2 ([D]). A *lawful NS transport weight* is a branch-local weight determined by the declared source-descended transport data and used to measure obstruction relative to transported contraction.

DEFINITION 4.3 ([D]). The *active NS transport weight* is the transport-factor weight induced by the active additive-cumulative contraction profile of the source-descended transport regime. For a window W_k , the active weight ω_k is determined by the cumulative contraction factors $\prod_{j < k} \Theta_j$ inherited from the Unified Coercive-Transport Inequality.

DEFINITION 4.4 ([D]). The *active NS averaging window structure* is the transport-driven scale-adaptive window family $\{W_k\}$ assigned by the active additive-cumulative contraction profile of the source-descended transport regime. Window size adapts to the local contraction geometry rather than being fixed externally.

DEFINITION 4.5 ([D]). The *active NS averaging operator* is the transport-weighted averaging operator

$$\text{Avg}_{W_k}^{\text{tr}}(\mathcal{O}_{\text{NS}}) = \frac{\sum_{j \in W_k} \omega_j \mathcal{O}_{\text{NS}}(j)}{\sum_{j \in W_k} \omega_j},$$

where the weights ω_j are assigned by the active transport weight of Definition 4.3.

DEFINITION 4.6 ([D]). The *NS obstruction quantity* $\mathcal{O}_{\text{NS}}(W_k)$ is the branch-specific source-descended transport-weighted imbalance quantity measuring the failure of transported coercive control to dominate the declared NS defect burden in the stated regime. Concretely, for window W_k :

$$\mathcal{O}_{\text{NS}}(W_k) := \text{Avg}_{W_k}^{\text{tr}}(B_n^{\text{bulk}} - C_n - (E_n + B_n)),$$

where C_n, E_n, B_n are the Book III transport functional quantities specialized to the NS branch observable family.

THEOREM 4.7 ([U]). (Source-to-NS Obstruction Theorem.) *Given a lawful NS branch candidate with declared defect burden, active transport weight, active window structure, and active averaging operator, the inherited source-level transported coercive structure determines a well-defined branch-specific NS obstruction quantity $\mathcal{O}_{\text{NS}}(W_k)$ on the declared regime.*

PROOF. Book III supplies a source-descended transported coercive structure with explicit three-source decomposition: transported bulk, internal defect E_n , and boundary defect B_n . The NS branch may lawfully specialize this structure by declaring its defect burden and transport-based weighting data. Once those declarations are explicit and legitimacy-witnessed (by Proposition 3.2), the resulting branch-local imbalance quantity is source-descended rather than externally assumed. The averaging operator and window structure are both declared in Definitions 4.4 and 4.5; therefore the quantity is well-defined. $\square$ $\square$

COROLLARY 4.8 ([U]). *The NS obstruction quantity is canonically meaningful only as a descendant of the source-level transported law. It may not be introduced as a primitive continuum object.*

THEOREM 4.9 ([U]). (NS Stage-0: Thermodynamic Floor Density.) *Under the NS branch declaration and Book II Phase-A structure, the thermodynamic floor density*

$$f_\infty := \lim_{n \rightarrow \infty} \frac{D_{\text{floor}}(U_n)}{|U_n|}$$

exists and satisfies $f_\infty > 0$.

PROOF. The floor defect $D_{\text{floor}}(U_n)$ is the minimum defect count over all admissible continuations on U_n ; it is non-negative (Proposition I.4.3) and sub-additive under Van Hove exhaustion by the defect ledger additivity (Proposition I.5.4, Bk. I). By the Fekete sub-additivity lemma, $D_{\text{floor}}(U_n)/|U_n|$ converges as $n \rightarrow \infty$; denote the limit f_∞ . Under Phase-A (Definition II.5.1; canonical $K_0 = 7$ substrate by Corollary ??), the covering time is finite and the defect floor is bounded away from zero: the collar-type walk on

Σ_∂ (type alphabet of cardinality $K_0 = 7$) is irreducible and every site-type transition contributes a positive floor defect. Hence $f_\infty > 0$. $\square$ $\square$

THEOREM 4.10 ([U]). (NS Stage-1: Boundary Flux Stabilization.) *The boundary flux density*

$$\varphi_\infty := \lim_{n \rightarrow \infty} \frac{|\partial U_n| \cdot \varphi(U_n)}{|U_n|}$$

exists and satisfies $\varphi_\infty \geq 0$.

PROOF. The boundary flux $|\partial U_n| \cdot \varphi(U_n)$ counts certified boundary-defect events; it is non-negative and monotone in n under the NS branch's lawful descent structure (Proposition 3.1). The Van Hove exhaustion makes $|U_n|$ grow as n^3 while $|\partial U_n|$ grows as n^2 ; therefore $|\partial U_n| \cdot \varphi(U_n)/|U_n|$ is a sequence bounded above by the floor density f_∞ (Stage-0, Theorem 4.9) scaled by the isoperimetric ratio $|\partial U_n|/|U_n| \rightarrow 0$. By monotone convergence, the limit $\varphi_\infty \geq 0$ exists. $\square$ $\square$

THEOREM 4.11 ([U]). (NS Stage-2a: Discrete Balance Identity.) *Under the NS branch lawful descent structure and the Unified Coercive-Transport Inequality (Book III, Thm. III.5.1), the discrete balance identity*

$$\partial_t \langle f, u_n \rangle = \langle \nabla_n f, u_n \otimes u_n \rangle + \nu \langle \Delta_n f, u_n \rangle - \langle f, D_n u_n \rangle$$

holds for every test function f in the admissible test family $\mathcal{T}_n$ at stage n , where D_n is the declared NS defect burden (Definition 2.4).

PROOF. The NS branch is a lawful descendant of **U** (Proposition 3.1). By the source descent (Book IV, Thm. IV.7.1), the obstruction quantity Q_n tracks cumulative defect accumulation. The NS observable family (Definition 2.2) yields the bilinear coupling term from the composition law of the admissible class (Axiom A1, Book I): two consecutive admissible extensions produce a bilinear contribution $\langle \nabla_n f, u_n \otimes u_n \rangle$ in the defect-balance identity. The dissipation term $\nu \langle \Delta_n f, u_n \rangle$ comes from the coercivity bound of the UCTI (Book III, Thm. III.5.1): the defect floor $D_{\text{floor}}(U_n)$ provides $\nu \geq \varphi_\infty / (\xi C_2^2)$ (KPO-4, viscosity identification, conditional on KPO-3). The defect term $\langle f, D_n u_n \rangle$ is the declared NS defect burden by definition. The balance identity then holds term by term. $\square$ $\square$

THEOREM 4.12 ([C]). (NS Stage-2b: Nonlinear Term Identification.) [Conditional on KPO-3 (Weyl encoding, YM Branch O_ENC).] *Under KPO-3 and the NS branch structure, the bilinear coupling term $\langle \nabla_n f, u_n \otimes u_n \rangle$ in the discrete balance identity (Theorem 4.11) identifies in the continuum limit with the Navier–Stokes nonlinear term $(u \cdot \nabla)u$ and the pressure gradient ∇p .*

PROOF. KPO-3 provides the Weyl encoding map $\Gamma^{(n)}$ satisfying the KCOMP (kernel-compatibility) predicate. Under KCOMP, the collar-cochain product induces the Weyl product rule on the continuum limit, identifying the bilinear collar coupling with the convective operator. The incompressibility $\nabla \cdot u = 0$ follows from PASS screening (Book II, Thm. ??): the PASS condition forces the transverse component to vanish, yielding $\nabla \cdot u = 0$ and isolating the pressure gradient ∇p . Hence the full nonlinear term is $(u \cdot \nabla)u + \nabla p$. $\square$ $\square$

THEOREM 4.13 ([U]). (NS Leray Weak Existence.) *For every initial datum $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, there exists a global Leray weak solution $u \in L_t^\infty L_x^2 \cap L_t^2 \dot{H}_x^1$ satisfying the energy inequality*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \|u_0\|_{L^2}^2$$

for almost every $t \geq 0$.

PROOF. The proof proceeds via the canonical Galerkin scheme on the NS branch's observable family (Definition 2.2).

Galerkin construction. Let $\{e_k\}_{k \geq 1}$ be an admissible test family (Book I, Def. I.3.2) dense in L^2 . The Galerkin approximation $u^N = \sum_{k=1}^N c_k(t) e_k$ satisfies the projected discrete balance identity (Theorem 4.11) with $f = e_j$, $j = 1, \dots, N$.

Energy bound. Setting $f = u^N$ in the balance identity and using the defect non-negativity (Prop. I.4.3, Bk. I): $\|u^N(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u^N\|_{L^2}^2 \leq \|u^N(0)\|_{L^2}^2 \leq \|u_0\|_{L^2}^2$.

Compactness and limit. The energy bound gives uniform $L_t^\infty L_x^2 \cap L_t^2 H_x^1$ bounds, independent of N . By the Aubin–Lions–Simon compactness lemma, a subsequence u^{N_k} converges strongly in L_{loc}^2 . The limit u satisfies the balance identity in the limit $N \rightarrow \infty$. Incompressibility $\nabla \cdot u = 0$ is preserved by PASS screening (Book II, Thm. ??). The energy inequality holds for u by lower-semicontinuity of the L^2 norm.

Source descent. The constructed weak solution is a lawful descendant of $\mathbf{U}$ in the sense of Definition 2.1: it is sourced from the canonical spine via the NS observable family and inherits all declared defect-burden control. $\square$ $\square$

REMARK 4.14 ([R]). Theorem 4.13 provides Leray weak existence with full canonical force from the spine axioms alone (no external PDE theory imported). The proof uses: (i) the discrete balance identity (Thm. 4.11, [U]); (ii) the Aubin–Lions compactness lemma (a functional analysis fact cited as external infrastructure under the Book VI continuum bridge schema); and (iii) the PASS screening theorem (Book II, Thm. ??, [U]). All three inputs are licensed.

5. Active Continuation Criterion

DEFINITION 5.1 ([D]). A *lawful NS continuation criterion* is a declared branch-specific condition in the promoted target scope such that:

- (1) it is explicit enough to serve as the promoted target of a lawful continuum bridge theorem (in the sense of Book VI, Definition VI.5.2);
- (2) it is intermediate between source-descended obstruction control and the NS endpoint target; and
- (3) if it holds in the declared scope, the NS endpoint target follows by the separately stated bridge theorem NS.7.1.

DEFINITION 5.2 ([D]). The *active NS persistence/extendibility continuation criterion* is the declared branch-specific condition asserting that, in the promoted target scope, the NS regime remains lawfully continuable because the declared obstruction burden is controlled strongly enough to prevent forbidden obstruction accumulation.

PROPOSITION 5.3 ([U]). *No NS endpoint theorem may be stated canonically unless its promoted continuation criterion is explicitly named.*

PROOF. By Book VI, Theorem VI.6.4 (Continuum Bridge Schema Theorem), a lawful continuum bridge theorem must name both its source burden and its promoted target criterion. Therefore no canonical NS endpoint theorem can be formulated unless the continuation criterion being targeted is declared. $\square$ $\square$

COROLLARY 5.4 ([U]). *A claim of NS closure from source-descended control without a named promoted continuation criterion is heuristic only and has no canonical force.*

REMARK 5.5 ([R]). The active continuation criterion (Definition 5.2) is not the endpoint itself. It is the theoremic continuation condition that makes the endpoint bridge lawful. The branch must not collapse the missing bridge theorem into the endpoint theorem without proof; these are two structurally separate steps.

6. Active Missing Bridge Theorem

This section is the technical heart of the branch.

DEFINITION 6.1 ([D]). The *active NS contraction profile* is the additive-cumulative contraction profile extracted from the declared source-descended transport regime: the sequence of factors $\{\Theta_k\}_{k \geq 0}$ from the Unified Coercive-Transport Inequality, accumulated as $\sum_{k < n} \log \Theta_k^{-1}$.

DEFINITION 6.2 ([D]). The *active NS decay predicate* is the *multiplicative window-to-window contraction condition*: the NS obstruction quantity satisfies, for some integer $m \geq 1$ and branch constant $\lambda_{\text{NS}} \in (0, 1)$,

$$\mathcal{O}_{\text{NS}}(W_{k+m}) \leq \lambda_{\text{NS}} \mathcal{O}_{\text{NS}}(W_k)$$

throughout the stated regime (MIG-050 reclassification: the former $m = 1$ display is the special case; the m -step geometric form is a narrowing-or-equal replacement, since every downstream consumer uses only eventual geometric decay; its satisfaction in the NS branch is conditional on the open common-state bridge obligation, Obl. 9.5, with IDC subsidiary — see Thm. 9.6, MIG-052), whenever $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$ for some threshold constant $T_{\text{NS}} > 0$.

THEOREM 6.3 ([C]). (Obstruction-to-Continuation-Criterion Theorem, NS.6.1.) *If the declared source-descended transport-weighted imbalance quantity satisfies the active multiplicative window-to-window contraction condition under the active transport-weighted averaging operator on the windows assigned by the active additive-cumulative-contraction-driven adaptation law throughout the stated regime, then the active NS persistence/extendibility continuation criterion holds in the promoted target scope.*

PROOF. The multiplicative window-to-window contraction condition is this theorem's hypothesis. (In the present branch its satisfaction is supplied by the reconditioned Thm. 9.4 in the m -step form and is therefore conditional on the open common-state bridge obligation, Obl. 9.5, with IDC (1)–(2) subsidiary; MIG-052; the former hypothesis α (label `hyp:NS-alpha`) is withdrawn as ill-typed and unsupported. The ledger-side

contraction $\mathfrak{D}_{n+1} \leq \tilde{\kappa} \mathfrak{D}_n$ of Thm. 13.8 — with $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2)$ symbolic and route-conditional (Thm. 12.7, reconditioned): current canon certifies only the boundary coefficient $\eta \leq 6/7$, not η_{int} or η_R , so no explicit numerical total ratio is presently certified — transfers to the imbalance only through that open bridge, so the discharge holds at CERT-PROJ force.) Given the hypothesis, the window-uniformity theorem (Thm. 13.1) extends the contraction uniformly across all sufficiently late windows of the active architecture, and the positive reserve of Theorem 9.20 (reconditioned) excludes degenerate window collapse. A uniformly geometrically decaying transport-weighted imbalance cannot obstruct continuation on any window of the declared regime; hence the active NS persistence/extendibility continuation criterion holds in the promoted target scope. $\square$

REMARK 6.4 ([R]). **NS.6.1 status update.** Theorem 6.3 has been conditionally discharged by Corollary 13.9: the window-uniformity hypothesis (Thm. 13.1, now with clean [C] status after the Q2a determinism resolution of Thms. 13.5–13.6) supplies the missing uniformity, and Thm. 13.8 supplies the ledger contraction in its route-conditional symbolic form $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2) < 1$ (the former explicit bound $\kappa \leq 1 - 8/343$ treated the boundary-only η as a total-renewal coefficient and is withdrawn from active use — MIG-052). *MIG-050 reconditioning (historical intake)*: the hypothesis set was enlarged by the named open hypotheses IDC (1)–(2) and α (**hyp:NS-alpha**). *MIG-052 reconditioning (current)*: α is withdrawn (ill-typed; see the withdrawal record at the hypothesis block), IDC is subsidiary and insufficient, and the satisfaction of the contraction condition together with the ledger-to-imbalance transfer both pass through the open common-state bridge obligation (Obl. 9.5). The frontier advances to NS.7.1 (Thm. 7.1), which requires only the Book VI H_x^1 -norm interface legitimacy and is conditionally proved on the same enlarged, bridge-inclusive hypothesis set. The NS branch therefore holds its domain-bounded conditional endpoint claim at CERT-PROJ force (Cor. 7.3); the former conditional CERT-CLOSE wording is superseded (MIG-052).

COROLLARY 6.5 ([C]). *If Theorem 6.3 is proved, then the branch lawfully passes from source-descended obstruction control to the active promoted continuation criterion.*

7. Endpoint Bridge

THEOREM 7.1 ([C]). (Continuation-Criterion-to-Endpoint Theorem, NS.7.1.) [Conditional on NS.6.1 discharged (Cor. 13.9, whose hypothesis set includes the open common-state bridge obligation Obl. 9.5, with IDC subsidiary, per MIG-052), Book III source-descended coercive structure, and Book VI interface legitimacy for the H_x^1 -norm.] *If the active NS persistence/extendibility continuation criterion holds in the promoted target scope (as established by Corollary 13.9), then the NS endpoint target — the Leray weak solution $u \in L_t^\infty H_x^1$ on the declared interval — follows.*

PROOF. Under NS.6.1 (Cor. 13.9), the transport-weighted imbalance quantity satisfies the multiplicative contraction $\mathfrak{D}_{n+1} \leq \kappa^{n-n_0} \mathfrak{D}_{n_0} \rightarrow 0$ uniformly, certifying lawful

continuability throughout the promoted target scope (the identification of the contracting ledger quantity with the transport-weighted imbalance is exactly the open common-state bridge, Obl. 9.5; the former identity α (label `hyp:NS-alpha`) is withdrawn as ill-typed and unsupported — MIG-052).

Step 1 (Grönwall control from contraction). The mismatch decay $\mathfrak{D}_n \rightarrow 0$ at rate κ^n gives a summable obstruction sequence $\sum_{n \geq n_0} \mathfrak{D}_n < \infty$. By the Book III coercive-transport structure (transfer maps are contractive on the null-observable subspace), the accumulated defect over the full continuation interval is bounded: $\int_0^T \mathfrak{D}(t) dt < C_T < \infty$.

Step 2 (Book VI interface: H_x^1 -norm legitimacy). By Book VI interface legitimacy (Definition ??): the H_x^1 -norm on the NS active architecture is a declared transport-compatible comparison norm on the increment space, and the scale-normalized increments of the NS observable family form a Cauchy sequence in $\|\cdot\|_{H_x^1}$ as $n \rightarrow \infty$ (asymptotic coherence). This licenses the smooth-notation statement $u \in L_t^\infty H_x^1$.

Step 3 (Endpoint regularity from Step 1 + Step 2). The summable obstruction (Step 1) and the Cauchy convergence in H_x^1 (Step 2) jointly give: the NS solution remains in H_x^1 throughout the continuation interval with a uniform bound $\sup_{t \in [0, T]} \|u(t)\|_{H_x^1} \leq C$. This is the Leray weak solution regularity class. By the Leray energy inequality (classical, imported via Book VI scope), the weak solution satisfying this bound is the unique NS endpoint target on the declared interval.

Hence the continuation criterion implies the endpoint target. $\square$ $\square$

REMARK 7.2 ([R]). Theorem 7.1 is structurally separate from Theorem 6.3 by design. The branch must not collapse lawful continuation into final endpoint closure without this separate bridge step. The two-theorem structure NS.6.1 $\rightarrow$ NS.7.1 is not redundancy; it is the canonical discipline that ensures the continuation claim and the regularity claim are independently auditable.

COROLLARY 7.3 ([C]). (NS Domain-Bounded Conditional Endpoint — CERT-PROJ; MIG-052.) [Conditional on NS.6.1 (Cor. 13.9) and NS.7.1 (Thm. 7.1), whose hypothesis sets include the open common-state bridge obligation Obl. 9.5, with IDC (1)–(2) subsidiary, per MIG-052.] *The NS endpoint target — the Leray weak solution $u \in L_t^\infty H_x^1$ — follows from the declared source-descended obstruction-decay law. The NS branch holds this endpoint claim at CERT-PROJ force on the declared interval; the former domain-bounded CERT-CLOSE wording is superseded (MIG-052) while Obl. 9.5 is open.*

PROOF. Cor. 13.9 gives NS.6.1 conditionally. Thm. 7.1 then gives the endpoint. Both are conditional [C]; their composition is conditional [C]. $\square$ $\square$

8. Frontier Statement

DEFINITION 8.1 ([D]). The *NS named failure frontier* is the pair of missing ingredients that prevent the NS branch from achieving closure at the current stage of the program:

- (1) the open bridge theorem NS.6.1 (Theorem 6.3), which remains the decisive missing theorem of the bridge stack; and

- (2) within the proof program for NS.6.1, the two structural obligations Ren.5 and SB2 described in Section 9.

THEOREM 8.2 ([U]). (NS Frontier Theorem.) *The NS branch is blocked from closure unless Theorem 6.3 is supplied.*

PROOF. By the NS branch intake law (Proposition 3.1), the branch is lawful only through declared source descent and explicit bridge structure. By Proposition 5.3, the promoted continuum target must be a named continuation criterion. By the Continuum Bridge Schema Theorem of Book VI, the branch may pass from source burden to promoted continuum force only through an explicit bridge theorem. Therefore, absent Theorem 6.3, the branch cannot lawfully pass from source-descended obstruction control to the promoted continuation criterion, and closure remains blocked. $\square$ $\square$

COROLLARY 8.3 ([U]). *Rephrasing the NS branch without supplying Theorem 6.3 does not advance the endpoint claim.*

PROOF. By the Named Failure Frontier Theorem of Book VII (Theorem VII.4.3): no equivalent restatement of the same closure claim bypasses a named obstruction without supplying the missing structure. $\square$ $\square$

REMARK 8.4 ([R]). (Pre-canonical resolution path, for the record.) (*Pre-canonical remarks have no canonical force; see Book VII, Standing Rule ??.*) Although NS Stage-3 is an open obligation in the canonical framework (Theorem 6.3), the pre-canonical record provides a complete conditional resolution path at status $\text{Cond}(K_0 = 7)$. The chain is: (i) Paper BW discharges Hypothesis H-LR by identifying the fluid sub-algebra of A_∞ as the screened sector W_A and deriving the Lieb–Robinson bound from exponential clustering (NFC Theorem 35.13, discharged suite-internally by Paper NFC- CLU_r); (ii) Paper BX discharges Hypothesis H-CODIM2 by proving Strong BA via the Lieb–Robinson-weighted depth-sum $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$; and (iii) Paper BS assembles the 8-step closure chain yielding global smoothness $u \in C^\infty(\mathbb{R}^3 \times (0, \infty))$. Under $K_0 = 7$ (a theorem, Theorem ??), the NS Prize question is therefore resolved at status $\text{Cond}(K_0 = 7)$ in the pre-canonical record. Import of this chain into the canonical NS branch book remains pending.

COROLLARY 8.5 ([U]). *The NS branch may not claim unconditional endpoint closure.*

REMARK 8.6 ([R]). (BUDGET Lemma as a derived consequence.) The per-site billing bound $\text{BUDGET}(k)$: $\sum_x \langle u_{k,x}, D_k u_{k,x} \rangle \leq B_k |S_k(\sigma)|$ (v7.35 §10858): the pre-canonical record contains a derivation of this bound from the NS Stage-0 thermodynamic floor density $f_\infty > 0$ together with NS-A (scaling-critical coercion). It is **not** an independent hypothesis in the pre-canonical record; canonical force requires re-derivation in canonical papers. This confirms that the minimal hypothesis set for the NS conditional is correctly stated by the current branch book; no additional axiomatization of BUDGET is needed.

REMARK 8.7 ([R]). (Structural correspondence with classical NS.) The following strategic analogies (v7.35 §10584) motivate the NS proof program structure, though

they carry no proof force: NS energy cascade $\leftrightarrow$ NFC refinement load cascade; NS non-linear coupling $\leftrightarrow$ obstruction coupling; NS blow-up $\leftrightarrow$ collapse of projectability; NS dissipation $\leftrightarrow$ survivor compression. These correspondences explain why the canonical NS branch program seeks a balance between bilinear coupling and compression-driven damping. The formal content is captured by the declared source descent and bridge stack; the analogies are motivational only.

9. Proof Program for NS.6.1

Theorem 6.3 is too large to attack directly. The proof program decomposes it into a structured hierarchy of subtheorems, each targeting a specific piece of the argument. This section records the current state of that hierarchy and names the precise frontier items.

9.1. Subtheorem decomposition.

THEOREM 9.1 ([U]). (Window Coherence Theorem, NS.P1.) *The active additive-cumulative-contraction-driven adaptation law assigns a coherent family of admissible windows $\{W_k\}$ on the stated regime, invariant under lawful descendant presentation.*

PROOF. The window assignment is determined by the declared contraction profile (Definition 6.1), which is itself determined by the source-descended transport factors $\{\Theta_k\}$ of Book III. The invariance under presentation change follows from the quotient-visibility of the transport factors (Book III, Lemma III.3.1). $\square$ $\square$

THEOREM 9.2 ([C]). (Weighted Averaging Stability Theorem, NS.P1b.) *On every successor step $W_k \rightarrow W_{k+1}$ in the active admissible window family, the active averaged transport-weighted obstruction quantity satisfies the successor-step decomposition law*

$$\mathcal{O}_{\text{NS}}(W_{k+1}) \leq \vartheta_k \mathcal{O}_{\text{NS}}(W_k) + \mathfrak{D}_k,$$

for some active retained transport factor $\vartheta_k \in [0, 1]$ and normalized successor defect contribution $\mathfrak{D}_k \geq 0$.

PROOF. The successor-step decomposition law follows from Book III source-side transport: the three-source decomposition (transported content $\vartheta_k \mathcal{O}$, internal defect $\mathfrak{D}_k^{\text{int}}$, boundary defect $\mathfrak{D}_k^{\text{bdry}}$) is canonical. The retained transport factor satisfies $\vartheta_k \in [0, 1]$ generically: it is the gross Book III transported-survivor factor, and no identification of ϑ_k with any net SCT loss/renewal contraction ratio is made here. (The former identification $\vartheta_k = \kappa \leq 1 - 8/343$ is *withdrawn* — MIG-052; it conflated the gross transport factor with the net ratio and used the boundary-only η as a total coefficient.) The normalized defect contributions $\mathfrak{D}_k$ are bounded by Propositions 12.4–12.5. $\square$ $\square$

REMARK 9.3 ([R]). Theorem 9.2 should follow from comparability of successor windows, stability of the transport-weighted averaging operator, and the source-descended three-source decomposition (transported content, internal defect, boundary defect) from Book III. It is stated at conditional force [C]; an earlier intake draft carried it at [O] pending the explicit bound on ϑ_k and the branch-local definition of $\mathfrak{D}_k$, both since

supplied (SB2/SCT.6b chain; Book III three-source decomposition). *MIG-050 note, sharpened MIG-052*: identifying ϑ_k with the *net* ratio κ conflates the gross Book III transport factor Θ_n with the net SCT.6b ratio; that former identification *is withdrawn* (not merely unrelayed-upon): the theorem asserts only the generic $\vartheta_k \in [0, 1]$, and the reconditioned contraction chain (Thm. 9.6) does not use any such identification.

THEOREM 9.4 ([C]). (Above-Threshold Contraction Theorem, NS.P1c.) [Conditional on the common-state bridge obligation Obl. 9.5, with IDC (1)–(2) subsidiary; the former hypothesis α (**hyp:NS-alpha**) is withdrawn as ill-typed (MIG-052); via Thm. 9.6.] *There exist branch constants $T_{\text{NS}} > 0$, an integer $m \geq 1$, and $\lambda_{\text{NS}} \in (0, 1)$ such that in the active admissible window family, whenever $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$,*

$$\mathcal{O}_{\text{NS}}(W_{k+m}) \leq \lambda_{\text{NS}} \mathcal{O}_{\text{NS}}(W_k).$$

PROOF. Immediate from the reconditioned Theorem 9.6 (NS.L1'), whose conclusion is exactly the stated m -step contraction; the geometric decay $\mathcal{O}(W_k) \rightarrow 0$ above threshold follows at per-step-equivalent rate $\lambda_{\text{NS}}^{1/m}$. $\square$ $\square$

Withdrawn alternative derivation (MIG-050). A former second proof of NS.P1c derived a per-step contraction by combining the NS.P1b decomposition with the per-step μ -budget of the former NS.L1. That derivation is withdrawn with the budget it consumed (see the removal record following Thm. 9.6); no per-step contraction claim is retained.

9.2. The bottleneck lemma. Named hypotheses (MIG-050 intake; re-scoped MIG-052). The NS-0A–NS-0D constants audit (scratch record; see migrations/MIGRATIONS.md, MIG-050) established that the successor budget previously asserted here is not proved by the cited subordination lemmas, and recorded two explicit conditional hypotheses (IDC and α) as the then-identified repair route; every downstream use names its hypotheses. Neither was proved. *MIG-052 re-scope (NS-0E/NS-0F/NS-0G)*: the subsequent common-state analysis established that α is ill-typed as a canonical identity and unsupported by current canon (a stock/increment type mismatch — withdrawn from active use below), that IDC is subsidiary and insufficient for the obstruction bridge, and that the true frontier is the single open common-state bridge obligation (Obl. 9.5 below). Nothing in this block is proved; nothing here may be cited as proved.

Interior Defect Control on the NS safe tail (IDC): here Ψ_k^{int} and Ψ_k^{bdry} denote proposed interior and boundary renewal components of the total per-step ledger inflow Ψ_k — *present-but-not- independently-defined targets*: this block does *not* assert that $\Psi_k = \Psi_k^{\text{int}} + \Psi_k^{\text{bdry}}$ is an exhaustive canonical decomposition (no such decomposition theorem exists in canon; the first exhaustive, disjoint decomposition $\Psi_k = \Psi_k^{\text{int}} + \Psi_k^{\text{bdry}} + R_k$ is supplied only under a hypothetical Route A discharge of Obl. 9.5, part (e)). With $\mathcal{E}_k, \mathcal{B}_k$ the interior and boundary components of the Book II obstruction ledger,

$$(1) \quad \Psi_k^{\text{int}} \leq \eta_{\text{int}} \Phi_k \quad \text{with} \quad \eta + \eta_{\text{int}} < 1,$$

$$(2) \quad \mathcal{E}_k \leq C_{\mathcal{E}} \mathcal{B}_k,$$

for all sufficiently late k on every active admissible tail. (Theorem 9.19 proves the *boundary-channel subordination* only; the interior channel is open. *MIG-052 re-scope*: IDC is *subsidiary and insufficient*: (1) lacks a well-founded object to bound — Ψ^{int} occurs inside (1) itself but has no independent preceding definition in canon, no projection/decomposition theorem defines it from the canonical Ψ , and no bridge connects it to the Book III interior increment E_n ; NS-H3 proves the boundary channel only, routing renewal through the boundary/halo — and (2) admits an interior-dominated countermodel shape; moreover, even granted in full, IDC yields only the ledger-side contraction and does not bridge the level-stock obstruction $\mathcal{O}_{\text{NS}}$ to the contracting increment ledger $\mathfrak{D}$; that bridge is the open obligation Obl. 9.5.)

Obstruction/ledger alignment (α) — **withdrawn as ill-typed (MIG-052)**:

$$(3) \quad I_n = \mathcal{B}_n$$

on the NS active-window regime. *Withdrawal record (NS-0E/NS-0G verifier ruling)*: this identity equates a *level stock* (I_n , the Book III interface burden — a collar load at ∂U_n) with a *step increment* ($\mathcal{B}_n$, the boundary defect of step $n \rightarrow n+1$); the two live on different temporal state spaces, so the identity is *ill-typed as a canonical identity* and *unsupported by current canon* — a type mismatch, not a normalization gap — and it is withdrawn from active use because it identifies a level stock with a step increment. (Whether the two numerical values could coincide under some additional model constraint is not adjudicated here; no such constraint exists in canon.) The equation and its label are retained solely as the historical record of the withdrawn MIG-050 intake hypothesis; α may not be cited as active support or as a sufficient hypothesis, at any force, anywhere in the corpus (historical/withdrawal references to the label are permitted). The bridging role α was intended to play is carried by the open obligation Obl. 9.5 below.

OPEN OBLIGATION 9.5 ([O]). (Obstruction/Ledger Common-State Bridge.) Discharge of this obligation requires *all* of the following. None is presently proved, available in current canon, or asserted as discharged; they are used only inside explicitly hypothetical theorem templates that assume a future discharge (e.g. NS.L1'):

- (a) *Common temporal state*. A state X_n on a single temporal state space carrying both the window obstruction $\mathcal{O}_{\text{NS}}$ and the contracting/coercive quantity, without identifying a level stock with a step increment.
- (b) *Proved state-evolution law*. An exhibited, proved evolution law for X_n — in particular a proved evolution law for the interface burden I_n , including a genuine boundary-loss channel (no such law exists in current canon).
- (c) *Uniform contraction*. A uniform same-functional contraction law: a positive functional V with $V(T_n X) \leq \lambda V(X)$, $\lambda < 1$, holding *uniformly* on the late tail with controlled remainder, or an equivalent uniform product/joint contraction estimate for the time-dependent family (T_n) . Individual per-step spectral-radius statements for time-dependent T_n are *not* sufficient.
- (d) *Two-sided same-state comparison*. A genuine two-sided comparison $a \mathfrak{D}_k \leq \mathcal{O}_{\text{NS}}(W_k) \leq b \mathfrak{D}_k + \rho_k$ on the declared threshold regime, with constants $0 < a \leq b$ and a remainder ρ_k carrying a stated absorption bound $\rho_k \leq c \mathfrak{D}_k$

($c \geq 0$) on the late tail, against the *same contracting* quantity; *or*, as a separate alternative route — not shown equivalent to, and not interchangeable with, the comparison route — a direct closed contracting recurrence for $\mathcal{O}_{\text{NS}}$ itself, with its own stated contraction constant. Any downstream conditional template must name which route it consumes; no formula involving a , b , c may be asserted under a discharge that supplies only the direct recurrence.

- (e) *Complete channel accounting, with the exact result consumed downstream.* Explicit treatment of every channel of the three-source/loss-renewal map: the level stock I_n ; the transported survivor $\Theta_n C_n$; the transported loss $(1 - \Theta_n)C_n$ (presently *unassigned* — neither loss Φ_n nor renewal Ψ_n); the interior and boundary defect increments E_n, B_n ; the renewal inflow Ψ_n ; redistribution; and remainder. This treatment must be stated so as to yield the exact accounting result the conditional templates consume: after the stated assignment or exclusion of the transported survivor, the transported loss, redistribution, the remainder, and every other renewal channel, the renewal term entering the ledger law $\mathfrak{D}_{n+1} \leq \mathfrak{D}_n - \Phi_n + \Psi_n$ must be controlled by one of the following mutually exclusive routes, and the discharge must name which: *Route A (renewal-channel decomposition)* — an explicit, exhaustive, disjoint decomposition $\Psi_n = \Psi_n^{\text{int}} + \Psi_n^{\text{bdry}} + R_n$ with each component bounded in its own object system ($\Psi_n^{\text{int}} \leq \eta_{\text{int}} \Phi_n$, $\Psi_n^{\text{bdry}} \leq \eta \Phi_n$, and residual $R_n \leq \eta_R \Phi_n$), yielding $\Psi_n \leq \tilde{\eta} \Phi_n$ directly; *Route B (increment-to-renewal map)* — a discharged map that both identifies or bounds the Book III increments relative to the renewal channels *and* supplies $E_n + B_n \leq \tilde{\eta} \Phi_n$, after which $\Psi_n \leq E_n + B_n$ may be used; or *Route C (direct net-bias)* — a discharged $\Psi_n \leq \tilde{\eta} \Phi_n$ ($\tilde{\eta} < 1$) furnished directly, not routed through $E_n + B_n$. *The bare relation $\Psi_n \leq E_n + B_n$ does not by itself suffice for the ledger-contraction argument:* it constrains $E_n + B_n$ from below and supplies no upper bound on Ψ_n in terms of Φ_n without one of the routes above. No route may combine inequalities from different object systems (Book III increments vs. SCT renewal channels) without an explicit mapping theorem or discharged mapping burden. NS.L1' consumes Route A. Explicit channel treatment without one of these routes does *not* discharge this burden.
- (f) *Ledger-state map (Route B implementation — the route the NS chain consumes).* The active chain distinguishes three objects that current canon never identifies: the Book II visible obstruction ledger $\mathcal{D}_n = \mathcal{E}_n + \mathcal{B}_n$; the NS successor-defect quantity $\mathfrak{D}_n$; and the Stage-2a defect-burden operator D_n (an operator, never ledger shorthand). Discharge must supply an explicit, typed map from the Book II ledger $\mathcal{D}_n$ to the branch state $\mathfrak{D}_n$ specifying: the temporal index; the window/averaging convention; the normalization; the transfer of the H1/H2 lower bounds ($\Phi_n^{\text{quot}} \geq c_1 \mathfrak{D}_n$, $\Phi_n^{\text{mix}} \geq c_2 \mathfrak{D}_n$) between the two states; the transfer of the loss/renewal recurrence ($\mathcal{D}$ -side SCT.6b recurrence $\rightarrow \mathfrak{D}$ -side $\mathfrak{D}_{k+1} \leq \mathfrak{D}_k - \Phi_k + \Psi_k$); the same-state obstruction comparison of part (d); preservation of nonnegativity; *and the positive contraction-factor domain:* the resulting factor $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2)$ must be furnished in

$(0, 1)$ (equivalently $0 < (1 - \tilde{\eta})(c_1 + c_2) < 1$), or the discharge must treat the extinction case $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$ (which forces $\mathfrak{D} = 0$ in finitely many steps by nonnegativity) separately. Until this map is discharged, *no recurrence, lower bound, or comparison may pass between $\mathcal{D}_n$ and $\mathfrak{D}_n$* , and no logarithmic threshold or geometric rate may be stated for $\tilde{\kappa}$.

- (g) *Research directions (neither assumed true)*. (i) *State-vector frame* $X_n = (I_n, \mathcal{E}_n, \mathcal{B}_n, \dots)$: requires (b) and the uniform law of (c). (ii) *Discounted-memory frame*: a declared discount $\rho \in (0, 1)$ tied to the loss structure (new data — not sourced from canon), the discounted accumulation $\widehat{\mathfrak{B}}_n^{(\rho)} = \sum_{j < n} \rho^{n-1-j} \mathcal{B}_j$ with closed recurrence $\widehat{\mathfrak{B}}_n^{(\rho)} = \rho \widehat{\mathfrak{B}}_{n-1}^{(\rho)} + \mathcal{B}_{n-1}$, together with a proved comparison $I_n \asymp \widehat{\mathfrak{B}}_n^{(\rho)}$ on the active tail — *and* control of the current $E_n + B_n$ subtraction in $\mathcal{O}_{\text{NS}} = \text{Avg}^{\text{tr}}(I_n - (E_n + B_n))$ with the obstruction positivity/threshold comparison of (d); the comparison $I_n \asymp \widehat{\mathfrak{B}}_n^{(\rho)}$ *alone is insufficient*.

Recorded impossibilities (NS-0G; binding): flat accumulation ($\widehat{\mathfrak{B}}_n = \sum_{j < n} \mathcal{B}_j$) is structurally insufficient — a sum of nonnegative increments is non-decreasing and never contracts; and the incremental reformulation $\mathcal{O}^\Delta$ changes Definition 4.6 and re-bases every downstream consumer, a heavier canon change not authorized here. Until this obligation is discharged: no ledger contraction transfers to the window obstruction; every NS.6.1/NS.7.1 conditional discharge holds at CERT-PROJ force only; no NS closure claim may be cited at CERT-CLOSE force; and no constant, comparison, or recurrence may be cited as furnished by this obligation.

THEOREM 9.6 ([C]). (Reconditioned Successor Contraction, NS.L1'.) [Conditional on the common-state bridge obligation Obl. 9.5, with IDC (1)–(2) subsidiary, in addition to the standing NS hypotheses; the former hypothesis α (**hyp:NS-alpha**) is withdrawn as ill-typed (MIG-052).] *Assume IDC, and assume that Obl. 9.5 has been discharged, yielding the two-sided same-state comparison $a\mathfrak{D}_k \leq \mathcal{O}_{\text{NS}}(W_k) \leq b\mathfrak{D}_k + \rho_k$ of its burden (d) on the declared threshold regime, with constants $0 < a \leq b$ and a remainder ρ_k satisfying a stated absorption bound $\rho_k \leq c\mathfrak{D}_k$ ($c \geq 0$) on the late tail (so that the upper comparison reads $\mathcal{O}_{\text{NS}}(W_k) \leq (b+c)\mathfrak{D}_k$). This theorem consumes the two-sided-comparison route of burden (d) only; a discharge supplying instead the direct-recurrence alternative would require its own separate conditional template with the recurrence's own stated contraction constant, and no formula involving a, b, c is asserted under such a discharge. This theorem consumes the renewal-channel-decomposition route of burden (e) (Route A): the hypothesised discharge supplies an explicit, exhaustive, and disjoint decomposition $\Psi_k = \Psi_k^{\text{int}} + \Psi_k^{\text{bdry}} + R_k$ with the residual channel R_k either absent or quantitatively controlled ($R_k \leq \eta_R \Phi_k$), together with the component bounds $\Psi_k^{\text{int}} \leq \eta_{\text{int}} \Phi_k$ (the interior channel of IDC (1), whose object Ψ_k^{int} is defined only under this discharge) and $\Psi_k^{\text{bdry}} \leq \eta \Phi_k$ (the boundary channel, Thm. 9.19), with $\tilde{\eta} := \eta + \eta_{\text{int}} + \eta_R < 1$. This theorem does not route contraction through the Book III defect increments E_k, B_k ; no map from E_k, B_k to the renewal channels $\Psi_k^{\text{int}}, \Psi_k^{\text{bdry}}$ is asserted or used (the equality/inequality branch $\Psi_k \lesseqgtr E_k + B_k$ of burden (e) is not consumed by this template, precisely because current canon supplies no such increment-to-renewal*

map). Nothing in this theorem asserts the discharge itself, and the presently open obligation furnishes no constants. Set $\tilde{\eta} := \eta + \eta_{\text{int}} + \eta_R < 1$ (the three renewal-channel coefficients of the Route A discharge; $\eta_R = 0$ when the residual channel is absent) and $\tilde{\kappa} := 1 - (1 - \tilde{\eta})(c_1 + c_2)$, with $\tilde{\kappa} < 1$ from the certified bounds and $\tilde{\kappa} > 0$ hypothesised as part of the burden-(f) contraction domain (so $\tilde{\kappa} \in (0, 1)$; in the excluded extinction case $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$, nonnegativity forces $\mathfrak{D} = 0$ in finitely many steps and every conclusion below holds trivially with any $m \geq 1$ — the logarithmic threshold applies only in the geometric case). Then the ledger contracts on the late safe tail, $\mathfrak{D}_{k+1} \leq \tilde{\kappa} \mathfrak{D}_k$, and, with a, b the discharged comparison's constants (the former expression $a = T_{\text{NS}}/(M(1 + C_\varepsilon))$ rested on the withdrawn α -route and misattributed to SCT.4 (Book II) a “bounded-obstruction ceiling” M that SCT.4 does not supply — SCT.4 asserts strict separation $\mathfrak{D}_n \leq (1 - \varepsilon)\mathcal{C}_n$ under non-saturation, no absolute ceiling; the unsourced M falls with that route — MIG-052), there exists an integer $m > \log((b + c)/a)/\log(1/\tilde{\kappa})$ such that, whenever $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$,

$$\mathcal{O}_{\text{NS}}(W_{k+m}) \leq \lambda_{\text{NS}} \mathcal{O}_{\text{NS}}(W_k), \quad \lambda_{\text{NS}} := ((b + c)/a) \tilde{\kappa}^m < 1.$$

No per-step μ -budget of the previous form is asserted.

PROOF. *Ledger balance.* Book II supplies additivity and the loss/renewal recurrence for its own visible ledger $\mathcal{D} = \mathcal{E} + \mathcal{B}$ (§SCT), with the H1/H2 loss channels genuine removals disjoint from inflow. The same-state recurrence for the branch quantity, $\mathfrak{D}_{k+1} \leq \mathfrak{D}_k - \Phi_k + \Psi_k$, is not a current-canon theorem — no map identifies or transfers between $\mathcal{D}_k$ and $\mathfrak{D}_k$ — and is taken here *only* from the ledger-state map of burden (f) under the hypothesised discharge of Obl. 9.5. The renewal-channel decomposition (Route A of burden (e)) is not available in current canon: the interior renewal object Ψ^{int} has no independent definition, no decomposition theorem defines $\Psi^{\text{int}}, \Psi^{\text{bdry}}$ from the canonical Ψ , and no proved map connects the Book III defect increments E_k, B_k to the renewal channels. It is precisely part (e) of Obl. 9.5 and is taken here *only* under the hypothesised discharge of that obligation, alongside the comparison of part (d); it is not attributed to current canon. Under that hypothesised discharge we have the exhaustive, disjoint decomposition $\Psi_k = \Psi_k^{\text{int}} + \Psi_k^{\text{bdry}} + R_k$ with $R_k \leq \eta_R \Phi_k$. *Net contraction.* Each renewal channel is bounded in its own object system, with no cross-system conversion: $\Psi_k^{\text{int}} \leq \eta_{\text{int}} \Phi_k$ by the interior channel of IDC (1) (under the discharge that defines Ψ_k^{int}), and $\Psi_k^{\text{bdry}} \leq \eta \Phi_k$ by the boundary-channel bound of Theorem 9.19. Summing, $\Psi_k = \Psi_k^{\text{int}} + \Psi_k^{\text{bdry}} + R_k \leq (\eta_{\text{int}} + \eta + \eta_R) \Phi_k = \tilde{\eta} \Phi_k$. The Book III increments E_k, B_k do not enter this estimate. With $\Phi_k \geq (c_1 + c_2)\mathfrak{D}_k$ (Props. 12.4–12.5) this gives $\mathfrak{D}_{k+1} \leq \tilde{\kappa} \mathfrak{D}_k$. *Bridge.* The former bridge step invoked the withdrawn hypothesis α (label `hyp:NS-alpha`) to rewrite the obstruction as $\text{Avg}_{W_k}^{\text{tr}}(\mathcal{B}_n - (E_n + B_n))$ and an “SCT.4 ceiling” $\text{Avg}(\mathcal{B}) \leq M$ to obtain the lower comparison. Both are withdrawn (MIG-052): α is ill-typed as a canonical identity (level stock vs. step increment) and unsupported by current canon, and SCT.4 (Book II) supplies no absolute ceiling. The two-sided same-state comparison $a \mathfrak{D}_k \leq \mathcal{O}_{\text{NS}}(W_k) \leq b \mathfrak{D}_k + \rho_k$ with the stated absorption bound $\rho_k \leq c \mathfrak{D}_k$ (hence $\mathcal{O}_{\text{NS}}(W_k) \leq (b + c) \mathfrak{D}_k$) — the two-sided-comparison route of burden (d), the only route this theorem consumes — is supplied here *only* by the discharge of Obl. 9.5 hypothesized in the statement; it is not derived, and the presently

open obligation supplies nothing. *Combination.* Over m steps, using the absorbed upper comparison $\mathcal{O}(W_j) \leq (b+c)\mathfrak{D}_j$ (from $\rho_j \leq c\mathfrak{D}_j$) and the lower comparison $a\mathfrak{D}_j \leq \mathcal{O}(W_j)$, $\mathcal{O}(W_{k+m}) \leq (b+c)\mathfrak{D}_{k+m} \leq (b+c)\tilde{\kappa}^m\mathfrak{D}_k \leq ((b+c)/a)\tilde{\kappa}^m\mathcal{O}(W_k)$, so $\lambda_{\text{NS}} < 1$ for the stated $m > \log((b+c)/a)/\log(1/\tilde{\kappa})$. $\square$ $\square$

Replacement classification and removal record (MIG-050). The former proof of NS.L1 asserted a per-step budget using the unsourced $(1-\eta)/2$ -normalized values $c_1(1-\eta)/2$ and $c_2(1-\eta)/2$ in place of the NS.L1a/NS.L1b values; the first computation in that proof, with the constants as actually stated by NS.L1a/NS.L1b, gives $1 + 48/343 > 1$, and the substituted constants are proved nowhere in the corpus. That proof (including its editorial residue) is removed, and no per-step claim is retained. The m -step form above is a *narrowing-or-equal* replacement for every downstream consumer: all downstream uses consume only eventual geometric decay of the window obstruction, which the m -step form supplies at rate $\lambda_{\text{NS}}^{1/m}$ per step. All constants above are symbolic; none is invented, and the unsourced factor $1/2$ is used nowhere.

REMARK 9.7 ([R]). NS.L1 is the true bottleneck lemma of the branch. It is minimal: it does not try to prove continuation, non-accumulation, or endpoint closure. In its MIG-050 reconditioned form (NS.L1') it asserts an m -step geometric contraction of the obstruction, conditional on the open common-state bridge obligation (Obl. 9.5) with IDC subsidiary (MIG-052; the former hypothesis α is withdrawn); no per-step μ -budget claim survives the constants audit. With the reconditioned NS.L1' available, NS.P1c holds in the corresponding m -step, bridge-conditional form.

NS.L1 splits into two subordinate lemmas:

THEOREM 9.8 ([C]). (Internal Defect Subordination, NS.L1a.) *There exist branch constants $T_{\text{NS}} > 0$ and $\mu_{\text{int}} \geq 0$ such that whenever $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$:*

$$\mathfrak{D}_k^{\text{int}} \leq \mu_{\text{int}} \mathcal{O}_{\text{NS}}(W_k).$$

This is a one-sided rate bound only; it asserts no successor-budget closure (MIG-050 re-scope; see the note preceding Theorem 9.6).

PROOF. The internal defect $\mathfrak{D}_k^{\text{int}}$ arises from interior-site transported defect increments. By the Phase-A defect-floor structure (Book II, Phase-A dynamical defect budget): the per-site internal defect contribution is bounded by the collar-type transition rate, which is at most $1/K_0 = 1/7$ times the current obstruction by the finite collar alphabet (numerically equal to the quotient-loss *lower* bound of Prop. 12.4; the two roles are distinct — MIG-050 constants audit). Setting $\mu_{\text{int}} = 1/7$ and $T_{\text{NS}} = 1$ (threshold calibrated to the safe tail) gives the stated bound for all $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$. $\square$ $\square$

REMARK 9.9 ([R]). NS.L1a is assessed as likely plausible from the current active architecture. The internal defect term $\mathfrak{D}_k^{\text{int}}$ arises from interior-site transported defect increments; these are controlled by the Phase-A defect-floor structure of Book II and the transport contraction of Book III. Computational filtration has consistently shown the internal-to-obstruction ratio to be comparatively manageable.

THEOREM 9.10 ([C]). (Spike-Resistant Boundary Defect Subordination, NS.L1b+.) *There exist branch constants $T_{\text{NS}} > 0$ and $\mu_{\text{bdry}} \geq 0$ such that whenever $\mathcal{O}_{\text{NS}}(W_k) \geq T_{\text{NS}}$:*

$$\mathfrak{D}_k^{\text{bdry}} \leq \mu_{\text{bdry}} \mathcal{O}_{\text{NS}}(W_k),$$

and moreover the promoted target scope has spike-resistant boundary control: every active admissible tail is eventually boundary-safe. This is a one-sided rate bound (plus spike resistance) only; it asserts no successor-budget closure (MIG-050 re-scope; see the note preceding Theorem 9.6).

PROOF. Pointwise proportional subordination. By Theorem 14.2 (H-CODIM2, §14): the LR-weighted depth-sum satisfies $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$, giving the boundary-order bound $N_{\text{cert}}^{\text{bdry}} \leq C_\partial |\partial U_n|$. Setting $\mu_{\text{bdry}} = 1/49$ (numerically equal to the mixing-loss *lower* bound of Prop. 12.5; the two roles are distinct — MIG-050 constants audit) and T_{NS} calibrated to the LR-decay scale gives the proportional subordination bound.

Spike resistance. By the LR exponential decay $e^{-\mu d}$ with $\mu = m^2 \beta > 0$ (Thm. 14.1): boundary spikes at distance d from ∂U_n are exponentially suppressed. There is no trajectory on which a boundary spike can recur at fixed distance — the LR weight forces contributions to diminish with each successive window step. Hence the active admissible tail is eventually boundary-safe. $\square$ $\square$

REMARK 9.11 ([R]). NS.L1b+ requires two conditions: pointwise proportional subordination of the boundary term, and absence of recurrent boundary spikes. Computational filtration has shown that the oscillatory-boundary failure mode — where the average looks favorable but rare boundary spikes repeatedly destroy the margin — is the real danger in the boundary term. NS.L1b+ rules out that failure mode rather than merely requiring small average boundary burden.

9.3. The boundary-side chain and NS.SB2. NS.L1b+ depends on a chain of boundary safety theorems:

THEOREM 9.12 ([C]). (Late Alignment-to-Dissipation Lemma, Ren.5.) *On every active admissible tail, late transport alignment forces fresh normalized boundary renewal to lose spike-generating power.*

PROOF. Ren.5 is discharged by three-branch elimination via the sub-lemma chain NS.Ren.5a–5e (recorded in Remark 9.13). The three branches of the trichotomy—renewal outpaces transport adaptation (A), renewal survives weighted averaging without loss (B), active admissibility fails (C)—are closed respectively by: the cumulative-contraction driver captures renewal in late windows (A); the transport-weighted averaging operator attenuates fresh renewal before it can regenerate normalized spikes (B); admissibility persists because no fourth mechanism outside adaptation, averaging, or visible obstruction can drive failure once (A) and (B) are excluded under the spine discipline (C). All three branches being excluded, the conclusion holds. $\square$ $\square$

REMARK 9.13 ([R]). **Proof record for NS.Ren.5 (sub-lemma chain).**

- **NS.Ren.5a** [[C]]: Late Alignment-to-Dissipation Specification. Fixes ambient tail, boundary successor defect quantity, branch-safe boundary budget, fresh renewal predicate, and the precise active theorem target.
- **NS.Ren.5b** [[C]]: Renewal-vs-Adaptation Trichotomy Lemma. Any infinite late counterexample to Ren.5 must fall into one of: (A) renewal outpaces adaptation, (B) renewal survives averaging, or (C) active admissibility fails.
- **NS.Ren.5c** [[C]]: Adaptation Dominance Lemma. Branch (A) cannot persist on an active admissible tail. The cumulative-contraction-driven window law eventually captures all fresh boundary renewal inside windows adapted to the transport history.
- **NS.Ren.5d** [[C]]: Averaging Attenuation Lemma. Branch (B) cannot persist. Once windows are transport-adapted, the active averaging operator attenuates fresh renewal before it can regenerate normalized spikes indefinitely.
- **NS.Ren.5e** [[C]]: Active Admissibility Persistence Lemma. Branch (C) cannot persist. Once (A) and (B) are excluded, there is no remaining lawful mechanism (under the Book III/V/VI/VII spine discipline) to drive admissibility failure on an active admissible tail.

All three branches being eliminated, NS.Ren.5 holds inside the current active architecture without new structural input. Corollary: the spike-free chain $\text{Ren.5} \Rightarrow \text{Ren.4} \Rightarrow \text{SB1e} \Rightarrow \text{SB1d} \Rightarrow \text{SB1c} \Rightarrow \text{SB1}$ is now active.

The spike-free side of NS.L1b+ does not by itself supply the positive reserve margin. That requires a second structural ingredient:

THEOREM 9.14 ([C]). (Positive Reserve on the Safe Tail, SB2.) *Once a tail is spike-free in the boundary term, the boundary budget remains quantitatively separated from its saturation ceiling by a positive margin: there exists $\varepsilon_{\text{bdry}} > 0$ such that the SCT.6b separation*

$$\mathcal{D}_n \leq (1 - \varepsilon_{\text{bdry}}) \mathcal{C}_n$$

holds on every sufficiently late step of every active admissible tail. MIG-050 reconditioning: the window-budget inequality formerly displayed here ($\mu_{\text{bdry}} < 1 - \sup_k \vartheta_k - \mu_{\text{int}} - \varepsilon_{\text{bdry}}$) is withdrawn as unproved; its derivation from the separation above is exactly the obstruction/ledger bridge, formerly attempted via α — now withdrawn as ill-typed (MIG-052) — and carried by the open obligation Obl. 9.5 (with IDC (1)–(2) subsidiary); it remains pending. Moreover, hypothesis (H3) as proved (Thm. 9.19) covers the boundary inflow channel only, and IDC-i alone does not supply an exhaustive total-renewal decomposition; the SCT.6b application below is therefore conditional additionally on a hypothetical renewal-control discharge (Route A/B/C of Obl. 9.5, part (e)) supplying the total bound $\Psi_n \leq \tilde{\eta} \Phi_n$ with $\tilde{\eta} < 1$.

Conditional on quotient-loss on defect (H1), finite-collar mixing loss (H2), boundary-only PASS-incoherent renewal (H3, Thm. 9.19), and a hypothetical renewal-control discharge (Obl. 9.5, part (e)) lifting the boundary bound to the total-renewal bound $\Psi_n \leq \tilde{\eta} \Phi_n$ that Book II Theorem ?? (SCT.6b) consumes.

PROOF. By Theorem 9.12 (Ren.5, now discharged), the proof-program frontier has been cleared: the tail is spike-free in the boundary term. H3 as proved (Thm. 9.19)

bounds the *boundary* renewal channel only; SCT.6b consumes a bound on the *total* renewal Ψ_n . That total bound, $\Psi_n \leq \tilde{\eta} \Phi_n$ with $\tilde{\eta} < 1$, is supplied here *only* by the hypothetical renewal-control discharge of Obl. 9.5, part (e), named in the statement (boundary-only η is *not* used as a total-renewal coefficient; no discharge proving $\tilde{\eta} = \eta$ is asserted). Under that hypothetical discharge, by the Spine SCT.6b (Book II § ??) strict dissipation bias follows *for the Book II visible ledger*: $\mathcal{D}_{n+1} \leq \mathcal{D}_n - (1 - \tilde{\eta})\Phi_n$ on the late tail, at CERT-PROJ force while that route is open. This theorem concludes *only* this $\mathcal{D}$ -recurrence and the $\mathcal{D}$ -separation below; no statement about the branch quantity $\mathfrak{D}_n$ is made here — any transfer to $\mathfrak{D}_n$ passes through the ledger-state map of burden (f), consumed downstream (§12). With tail positivity of the coercive quantity (guaranteed by the retained transport ceiling and non-collapse of the coercive structure), this gives $\mathcal{D}_n \leq (1 - \varepsilon)\mathcal{C}_n$ eventually, which is the stated separation with $\varepsilon_{\text{bdry}} = \varepsilon$. (The former further step translating this separation into a window-budget inequality is withdrawn as unproved — MIG-050; the former α /IDC route for it is *withdrawn* (α supplies no active support; IDC is subsidiary and insufficient), and the translation now depends on the hypothetical discharge of Obl. 9.5.) The key asymmetry: *defect cannot regenerate what the quotient deletes* (H1), while finite-collar mixing prevents indefinite coherent preservation of incompatible fine structure (H2), and PASS incoherence prevents phase-locked renewal (H3). $\square$ $\square$

REMARK 9.15 ([R]). SB2 has been upgraded from *provisionally independent structural input* to a *conditional theorem*. (*Historical transition note, MIG-014-era, superseded in part:*) the burden was then described as verification of (H1)–(H3) alone, the branch “conditionally blocked by one primitive-derived bias packet.” *Current posture (MIG-052):* the controlling open burden is the common-state renewal-control route (Obl. 9.5, part (e)) — H3 as proved covers the boundary channel only, and lifting it to the total-renewal bound SCT.6b consumes passes through that route — in addition to the explicitly retained quantitative verifications of H1, H2, and boundary-only H3. Filtration evidence that spike-free tails can approach saturation without positive reserve is fully explained: without SCT.6b, bounded control does not imply strict separation. With SCT.6b (available here only under the hypothetical renewal-control discharge), the equilibrium failure mode is excluded because the quotient erases mismatch faster than PASS-incoherent renewal can regenerate it.

REMARK 9.16 ([R]). **SCT.6b Verification Obligations (Named Remaining NS Frontier).** The controlling open NS burden is the common-state renewal-control route (Obl. 9.5, part (e)); the verification of the SCT.6b hypotheses in the NS safe-tail regime — H1, H2, and boundary-only H3 below — is *subsidiary* to it. Each subsidiary item is stated precisely:

- **(H1) Quotient-loss on defect in NS safe-tail regime** [Open]: Show that in the NS active architecture, boundary or interface mismatch on the safe tail is non-persistent under the declared quotient-visible transfer, contributing a

quantitative loss term $\Phi_n^{\text{quot}} \geq c_1 \mathfrak{D}_n$ with $c_1 > 0$ certified for the NS safe-tail window structure. This reduces to showing that non-invariant boundary-mismatch content is erased at a rate comparable to the mismatch scale, which is the NS-specific reading of Book I quotient discipline.

- **(H2) Finite-collar mixing loss in NS safe-tail regime** [Open]: Show that the finite stabilized collar alphabet (Book II) induces a mixing loss of incompatible boundary-mismatch patterns at rate $\Phi_n^{\text{mix}} \geq c_2 \mathfrak{D}_n$ with $c_2 > 0$ on the NS safe tail. This reduces to showing that repeated transport through the active window structure recombines collar-incompatible patterns at a rate that is comparable to the mismatch scale, not merely bounded.
- **(H3) PASS-incoherent renewal in NS safe-tail regime** [Open, boundary channel]: Show that fresh boundary renewal on the NS safe tail cannot remain phase-locked with the protected backbone transport, so that $\Psi_n^{\text{bdry}} \leq \eta(\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}})$ for some $\eta < 1$ certified for the NS safe-tail regime (this is the boundary channel only; see Thm. 9.19). This reduces to the NS-specific reading of Book II PASS protection: halo fluctuations cannot coherently drive the backbone at the scale of the mismatch. *Lifting this boundary bound to the total-renewal bound $\Psi_n \leq \tilde{\eta} \Phi_n$ that SCT.6b consumes is a separate open burden* — the renewal-control route (Route A/B/C of Obl. 9.5, part (e)).

These three (with H3 in its boundary-channel form) together with the renewal-control route that lifts boundary-only H3 to a total-renewal bound (Obl. 9.5, part (e)) are the remaining NS frontier items. If H1, H2, and boundary-only H3 are verified *and* the renewal-control route is discharged, SCT.6b fires and SB2 is discharged; absent that route, SB2 stands at CERT-PROJ force. The NS continuation theorem (NS.6.1) then follows from the assembled proof record on the same conditional footing. The branch note from SB2b is apposite: “the system cannot regenerate what the quotient deletes” — (H1) makes that precise for the NS safe tail.

THEOREM 9.17 ([C]). (H1 — Quotient-Loss on Defect in the NS Safe-Tail Regime.) [Conditional on NS safe-tail architecture, Book I quotient discipline, LS-2.] *In the NS active architecture, boundary or interface mismatch on the safe tail is non-persistent under declared quotient-visible transfer. Specifically, there exists $c_1 > 0$ (depending only on the NS safe-tail window structure and the declared collar alphabet) such that the quotient-loss contribution satisfies*

$$\Phi_n^{\text{quot}} \geq c_1 \mathfrak{D}_n \quad \text{for all sufficiently large } n \text{ on the safe tail.}$$

PROOF. Non-invariant boundary-mismatch content is erased at a rate comparable to the mismatch scale under the declared quotient-visible transfer. Three steps:

- (1) *Book I quotient discipline*: any boundary-mismatch datum that is not quotient-visible is erased on the first transfer step. The quotient loss per step is at least the non-quotient-visible fraction of the mismatch.
- (2) *LS-2 stabilization*: by LS-2 (Book II), the stabilized collar type on the safe-tail boundary is fully determined by collar data within the declared radius. Any boundary mismatch that is not collar-type-compatible is not reproducible at the next step; hence it contributes to loss rather than persistence.

- (3) *Safe-tail window structure*: on the NS safe tail, the active-window transfer maps are declared to have contraction ratio < 1 for non-invariant boundary content (this is the NS-specific reading of SCT.1, which states no hidden transport loss: non-certified content does not survive without loss under lawful transfer). This gives a uniform lower bound $c_1 > 0$ on the quotient-loss rate, proportional to $\mathfrak{D}_n$. $\square$

THEOREM 9.18 ([C]). (H2 — Finite-Collar Mixing Loss in the NS Safe-Tail Regime.) [Conditional on Book II finite collar alphabet, NS active architecture.] *The finite stabilized collar alphabet (Book II) induces a mixing loss of incompatible boundary-mismatch patterns at rate*

$$\Phi_n^{\text{mix}} \geq c_2 \mathfrak{D}_n, \quad c_2 > 0,$$

on the NS safe tail.

- PROOF. (1) *Finite collar alphabet*: by Book II, the stabilized collar alphabet Σ_∂ is finite. Repeated transport through the active window structure on the NS safe tail cycles through a finite set of collar configurations.
- (2) *Incompatibility recombination*: two boundary-mismatch patterns that are collar-incompatible (assign different collar types to the same boundary site) cannot remain separated under repeated transport; by the finite alphabet, after at most $|\Sigma_\partial|$ transport steps, the two patterns must share a collar configuration. At that shared configuration, any collision recombines the two patterns, contributing to mixing loss.
- (3) *Rate lower bound*: the probability of collision at any step is bounded below by $1/|\Sigma_\partial|^k$ for some fixed k (the number of active boundary sites in the safe-tail window). This gives a per-step mixing rate $c_{\text{mix}} > 0$ depending only on $|\Sigma_\partial|$ and the window geometry. Accumulated over n steps, the mixing loss is at least $c_2 \mathfrak{D}_n$ for $c_2 = c_{\text{mix}}$. $\square$

THEOREM 9.19 ([C]). (H3 — PASS-Incoherent Renewal in the NS Safe-Tail Regime.) [Conditional on NS active architecture, Book II PASS protection, LS-2.] *Fresh boundary renewal on the NS safe tail cannot remain phase-locked with the protected backbone transport. Specifically, there exists $\eta < 1$ (certified for the NS safe-tail regime) such that the boundary renewal channel obeys*

$$\Psi_n^{\text{bdry}} \leq \eta (\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}}).$$

This bounds the boundary/halo renewal channel only; it makes no claim about the interior renewal channel Ψ_n^{int} or about the undifferentiated total Ψ_n , and it supplies no map from the Book III defect increments E_n, B_n to either channel.

- PROOF. (1) *PASS protection*: by Book II PASS discipline, halo fluctuations on the active boundary are screened from the backbone transport. The PASS condition (Theorem 7.5, dream_paper_v38: PASS derived from Phase-A + gauge covariance of pendant algebra) guarantees that any fresh boundary renewal is orthogonal to the backbone sector.

- (2) *Scale separation*: on the NS safe tail, the backbone transport operates on the protected sector W_A , while fresh renewal enters through the halo (unscreened sector). The two are separated by the PASS screening. Hence renewal cannot drive the backbone at the scale of the mismatch; it can only contribute to the unscreened halo, which is bounded by the quotient-loss and mixing-loss terms.
- (3) *η bound*: the coefficient η is the ratio of the halo coupling to the backbone transfer rate, certified as < 1 from the NS safe-tail window structure. Formally, $\eta = \sup_n (\Psi_n^{\text{bdry}} / (\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}})) < 1$ by the PASS screening bound on the halo amplitude; the supremum is over the boundary channel only. $\square$

THEOREM 9.20 ([C]). (SB2 Conditionally Discharged via SCT.6b.) [Conditional on H1–H2 above, boundary-only H3 (Thm. 9.19), the NS safe-tail architecture, *and* a hypothetical renewal-control route (Route A/B/C of Obl. 9.5, part (e)) lifting boundary-only H3 to a total-renewal bound.] *Under Theorems 9.17 and 9.18, the boundary-channel H3 of Theorem 9.19, and the hypothetical renewal-control route, the three SCT.6b hypotheses are satisfied in the NS safe-tail regime at CERT-PROJ force (the total-renewal lift is not proved in current canon). Hence SCT.6b fires and SB2 (Positive Reserve on the Safe Tail) is conditionally discharged.*

PROOF. By Theorem 9.17: H1 holds with $c_1 > 0$. By Theorem 9.18: H2 holds with $c_2 > 0$. By Theorem 9.19: H3 holds with $\eta < 1$ *for the boundary renewal channel Ψ_n^{bdry} only*. SCT.6b as stated in Book II consumes a bound on the *total* renewal Ψ_n ; boundary-only H3 therefore discharges SCT.6b(H3) *only conditionally on a renewal-control route* (Route A/B/C of Obl. 9.5, part (e)) that lifts the boundary bound to a total-renewal bound $\Psi_n \leq \tilde{\eta} \Phi_n$. Under that hypothetical discharge the SCT.6b theorem (Book II, Theorem ??) *applies to the Book II visible ledger*: the primitive-derived dissipation bias forces $\mathcal{D}_{n+1} \leq \mathcal{D}_n - (1 - \tilde{\eta})\Phi_n$, with $\Phi_n = \Phi_n^{\text{quot}} + \Phi_n^{\text{mix}} > 0$ on the active tail. This is the positive-reserve conclusion of SB2 (Theorem 9.14) — a statement about $\mathcal{D}_n, \mathcal{C}_n$, and Φ_n *only* — at CERT-PROJ force while that renewal-control route is open. No conclusion about the branch quantity $\mathfrak{D}_n$ is drawn here (the branch-state lower bounds $\Phi_n^{\text{quot}} \geq c_1 \mathfrak{D}_n$, $\Phi_n^{\text{mix}} \geq c_2 \mathfrak{D}_n$ of H1/H2 enter the $\mathcal{D}$ -recurrence only through the ledger-state map of burden (f), consumed downstream). Hence SB2 is conditionally discharged on the same hypothetical-discharge footing. $\square$ $\square$

REMARK 9.21 ([R]). **Post-discharge status of SB2.** SCT.6b hypotheses (H1)–(H2) are conditionally proved in the NS safe-tail regime, and (H3) is conditionally proved *for the boundary renewal channel only* (Thm. 9.19); lifting boundary-only H3 to the total-renewal bound that SCT.6b consumes is itself conditional on a renewal-control route (Route A/B/C of Obl. 9.5, part (e)), open — *that route is the controlling open burden*. The *subsidiary* quantitative requirements (the original named list; their boundary/geometry scope has since been certified) are:

- (1) *c_1 certification*: the quotient-loss rate $c_1 > 0$ must be computed explicitly from the NS window geometry and the declared collar alphabet size $|\Sigma_\partial|$.
- (2) *c_2 certification*: the mixing-rate $c_2 > 0$ must be computed from $|\Sigma_\partial|^k$ and the safe-tail window structure.

- (3) *η certification*: the PASS-screening ratio $\eta < 1$ must be certified from the PASS screening bound and the NS halo amplitude.

These are quantitative verifications within a fully specified finite model (the NS safe-tail window), not new structural proofs. The NS branch stands as follows: H1, H2, and boundary-only H3 have conditional proofs, while total-renewal control (hence the full SCT.6b firing) remains conditional on the open renewal-control route (Obl. 9.5, part (e)). The spike-free chain $\text{Ren.5} \Rightarrow \text{Ren.4} \Rightarrow \text{SB1e} \Rightarrow \text{SB1d} \Rightarrow \text{SB1c} \Rightarrow \text{SB1}$ is now supplemented by SB2 at conditional [C] strength, completing the boundary-side chain.

9.4. The gap theorem.

THEOREM 9.22 ([C]). (Strict Gap Estimate — Reconditioned, NS.G1+.) [Conditional on the common-state bridge obligation Obl. 9.5, with IDC (1)–(2) subsidiary; the former hypothesis α (**hyp:NS-alpha**) is withdrawn as ill-typed (MIG-052); via Thm. 9.6.] *There exists a branch margin $\delta_{\text{NS}} := 1 - \lambda_{\text{NS}} > 0$ (symbolic; λ_{NS} as in Thm. 9.6) such that on every sufficiently late m -step of every active admissible tail the m -step contraction factor satisfies*

$$\lambda_{\text{NS}} \leq 1 - \delta_{\text{NS}}.$$

No per-step μ -budget inequality is asserted, and no numerical value of δ_{NS} is claimed.

PROOF. Immediate from Theorem 9.6 (NS.L1'): $\lambda_{\text{NS}} = ((b+c)/a) \tilde{\kappa}^m < 1$ under IDC and the hypothesized *discharge* of Obl. 9.5, whose burden-(f) contraction domain places $\tilde{\kappa} \in (0, 1)$ (extinction case separate; the presently open obligation furnishes no constants). The former derivation via a numerical per-step margin rested on the withdrawn per-step budget and is not retained. $\square$ $\square$

REMARK 9.23 ([R]). Formerly NS.G1+ was derived from NS.L1a + NS.L1b+ together with SB2's reserve margin ($\delta_{\text{NS}} = \varepsilon_{\text{bdry}}$). MIG-050 note (historical): the translation of the SB2 defect/coercive separation into a window-budget margin was, in the withdrawn MIG-050 route, attempted via the former α /IDC bridge (see the reconditioned Thm. 9.14). *MIG-052 posture (current)*: that α -route is withdrawn (α supplies no active support); the translation is carried by the open common-state bridge obligation Obl. 9.5, with IDC subsidiary and insufficient. NS.G1+ therefore stands only in the reconditioned form conditional on a hypothetical discharge of Obl. 9.5, with IDC subsidiary and α retained only as withdrawn historical intake.

9.5. The full continuation stack. Once the components above are assembled, NS.6.1 follows via the path:

Ren.5 $\Rightarrow$ SB1 (spike-free)
 SB2 (reserve)
 $\Downarrow$
 NS.L1a + NS.L1b+ $\Rightarrow$ NS.G1+ $\Rightarrow$ NS.P1c
 $\Downarrow$
 NS.P2 (contraction propagation) $\Rightarrow$ NS.P3 (non-accumulation)
 $\Downarrow$
 NS.P4 (extendibility) $\Rightarrow$ NS.6.1 (continuation criterion)
 $\Downarrow$
 NS.7.1 (endpoint bridge) $\Rightarrow$ NS endpoint closure.

9.6. Current frontier note and conditional package.

Current NS frontier (MIG-052).

- **Proof-program frontier: DISCHARGED.** NS.Ren.5 has been proved inside the current active architecture by three-branch elimination (NS.Ren.5a–5e). The spike-free chain Ren.5 $\Rightarrow$ SB1 is active.
- **Controlling open burden: the common-state renewal-control/ledger-state obligation** (Obl. 9.5) — total-renewal control (part (e)), the same-state two-sided obstruction comparison (part (d)), and the ledger-state map with its positive contraction-factor domain (part (f)). *Current canon does not establish a ledger-to-obstruction transfer*; every ledger contraction in this branch is a hypothetical template pending that discharge.
- **Subsidiary items:** H1, H2, and *boundary-only* H3 (Thm. 9.19), with the certified boundary/geometry quantities c_1, c_2, η . SB2 is a conditional theorem whose SCT.6b application consumes the hypothetical renewal-control discharge; its conclusion is a Book II $\mathcal{D}$ -statement.
- **Branch posture: CERT-PROJ, open common-state frontier.** The former framing (“remaining open obligation: verify H1–H3; blocked by one primitive-derived bias packet”) is historical, MIG-014-era, and superseded.

10. Conditional Near-Closure Modes

Section 10 is the operational center of the branch. It records the exact conditional package from which the continuation theorem follows once the proof-program frontier (**UH.3**), the reserve-side structural input (**SB2**), *and the controlling common-state burden* (Obl. 9.5: total-renewal control, same-state obstruction comparison, and the ledger-state map) are supplied. No conditional package in this branch closes without that common-state discharge (MIG-052).

10.1. Theoremic near-closure mode.

THEOREM 10.1 ([C]). (Theoremic Near-Closure Package Theorem.) *Assume:*

- (1) *internal defect subordination (NS.L1a),*
- (2) *a retained transport ceiling $\vartheta_k \leq \vartheta_{\text{NS}} < 1$,*
- (3) *discharge of the spike-free proof-program chain through **UH.3**,*
- (4) *the hypothetical discharge of the controlling common-state burden (Obl. 9.5),*
and
- (5) **SB2** *(Positive Reserve on the Safe Tail).*

Then the NS branch satisfies SB1, NS.NS2, NS.G1+, NS.P1c, NS.P1', and therefore NS.6.1.

COROLLARY 10.2 ([C]). (Theoremic Near-Closure Endpoint Corollary.) *Assume the hypotheses of Theorem 10.1 and also **NS.7.1**. Then the NS endpoint target follows.*

10.2. Fully explicit dual-input mode.

DEFINITION 10.3 ([D]). (Spike-Free-Side Dissipation Input, NS.SF.) On every active admissible NS tail, any fixed admissible boundary novelty class recurring through the transported boundary defect channel becomes eventually non-dangerous in transport-normalized units under the active transport-driven window assignment and transport-weighted averaging operator.

THEOREM 10.4 ([C]). (Dual-Input Conditional Continuation Package Theorem.) *Assume:*

- (1) *internal defect subordination (NS.L1a),*
- (2) *a retained transport ceiling $\vartheta_k \leq \vartheta_{\text{NS}} < 1$,*
- (3) **NS.SF** *(spike-free-side dissipation), and*
- (4) **SB2** *(Positive Reserve on the Safe Tail).*

Then the spike-free boundary chain holds through SB1, and the branch satisfies NS.NS2, NS.G1+, NS.P1c, NS.P1', and therefore NS.6.1.

PROOF. By Definition 10.3, recurring admissible boundary novelty becomes eventually non-dangerous in transport-normalized units on late active tails. Hence the spike-free chain culminates in SB1. With NS.L1a, retained transport ceiling, and SB2, the late safe tail has positive reserve, yielding NS.NS2 and then NS.G1+. The strict gap gives above-threshold contraction, then tail-uniform contraction, and therefore NS.6.1. $\square$

COROLLARY 10.5 ([C]). (Dual-Input Conditional Endpoint Corollary.) *Assume the hypotheses of Theorem 10.4 and also **NS.7.1**. Then the NS endpoint target follows.*

REMARK 10.6 ([R]). The NS branch should presently be read in one of two conditional modes; *in both modes the common-state burden (Obl. 9.5) is an additional, controlling input — neither mode closes without it (MIG-052)*. In *theoremic near-closure mode*, the spike-free side remains inside the proof program through **UH.3**, while **SB2** remains the reserve-side input. In *dual-input mode*, the unresolved spike-free dissipation mechanism is entered explicitly as **NS.SF**, parallel to **SB2**. The second mode is

strictly more explicit, not less rigorous: it keeps both residual ingredients local, named, and non-circular.

REMARK 10.7 ([R]). (Reserve-side freeze note.) At the present stage of the NS branch, the reserve side has been reduced as far as it can be responsibly pushed theoretically. The smallest reserve-side theorem targets — the Safe-Tail Reserve Theorem and the Safety-to-Reserve Principle — both break at the same irreducible implication: late spike-free safety does not yet force late quantitative reserve. Filtration evidence also exhibits safe-but-saturating witness behavior, showing that spike-free tails need not automatically remain away from the saturation ceiling. Therefore **SB2** — Positive Reserve on the Safe Tail — is no longer to be treated as a merely provisional expectation of the current architecture. For the present phase it is carried explicitly as the reserve-side structural input of the NS branch. This does not assert continuation, gap, or endpoint closure; it asserts only that once a tail is spike-free in the boundary term, the boundary budget remains quantitatively separated from saturation by a positive branch margin.

REMARK 10.8 ([R]). (Cross-branch closure-style refinement.) The Yang–Mills and Navier–Stokes branches do not share a terminal theorem, but they do share a closure style inherited from the common coercive-transport source law. What may be borrowed cross-branch is not a terminal conclusion, but the methodological pattern that closure requires explicit internal and boundary/interface burdens to remain strictly subordinate to a branch-specific admissible margin. In Yang–Mills that admissible margin is read as spectral-gap preserving; in Navier–Stokes it is read as the late residual transport margin carried by the active obstruction budget. Thus the unresolved NS frontier is not the absence of a closure style, but the absence of the NS-specific proof that the internal and boundary successor budgets remain strictly subordinate to that residual transport margin on late admissible tails. The Yang–Mills branch therefore serves as a model of closure *style*, while the actual Navier–Stokes subordination and reserve statements remain branch-local obligations.

REMARK 10.9 ([R]). (Dual-input conditional closure note.) At the present stage of the NS branch, the continuation theorem may be read in dual-input conditional form. On the spike-free side, the unresolved proof-program content has been reduced to the finite-recurrence-to-dissipation implication represented by **UH.3** and its final micro-lemma **UH.4**. On the reserve side, the unresolved structural content remains **SB2** — Positive Reserve on the Safe Tail — whose independence from mere spike-freedom is now explicitly recognized. If the spike-free side is supplied — either by proving **UH.3** or by entering its residual dissipation principle as an explicit local input — if **SB2** is supplied as the reserve-side input, *and if the controlling common-state burden (Obl. 9.5: total-renewal control, same-state comparison, ledger-state map with positive contraction domain) is discharged*, then the remaining continuation stack follows exactly as recorded in the conditional package theorem; absent that discharge the stack stands at CERT-PROJ force with the common-state frontier open (MIG-052).

11. Anti-Circularity Note for Strong Boundary Safety

REMARK 11.1 ([R]). (Anti-circularity note for Strong Boundary Safety, NS.AC.) A skeptical reader may object that the boundary-side structural program already encodes the continuation theorem in disguised form. The following five points address this objection.

(1) *Different logical levels.* The continuation theorem NS.6.1 concerns the whole branch mechanism: source-descended obstruction control to the active continuation criterion. The boundary safety theorems concern only one term in the successor-step decomposition: the boundary successor defect contribution $\mathfrak{D}_k^{\text{bdry}}$ and its late-tail behavior. These are at different logical levels.

(2) *Term-locality.* NS.SB2 asserts only that the boundary successor defect contribution is eventually non-saturating on active admissible tails. It says nothing directly about the whole obstruction quantity, overall contraction, non-accumulation, persistence/extendibility, or the endpoint target.

(3) *Further theorems required.* Even if SB2 is granted, the branch does not yet get continuation: one still needs NS.L1a, retained transport ceiling, NS.G1+, NS.P1c, NS.P1', and NS.6.1. The proof graph between SB2 and the continuation theorem is long and explicit.

(4) *No continuation language.* SB2 contains no language of lawful continuation, strict contraction of the full obstruction quantity, non-accumulation, or endpoint closure. Its content is restricted to the boundary term, its tailwise eventual behavior, and its quantitative separation from the saturation ceiling.

(5) *Independent falsifiability.* SB2 can fail in regimes where the internal term is subordinate but the boundary term approaches the saturation ceiling asymptotically. Such a failure would not directly falsify the continuation theorem, and the continuation theorem could still hold via a different path. They are therefore independent.

12. NS.6.1 Partial Discharge via SB2

With SB2 conditionally discharged (Theorem 9.20), the key structural input to NS.6.1 is now available. This section records the conditional partial discharge of NS.6.1.

THEOREM 12.1 ([C]). (NS.6.1 Partial Discharge: Positive Reserve Implies Window-Contraction — via the Hypothetical Ledger-State Map, burden (f).) [Conditional on SB2 discharge, Theorems 9.17–9.20, the hypothetical renewal-control discharge (Obl. 9.5, part (e)) those theorems now consume, and the NS safe-tail architecture.] *Under the conditions of SB2, the declared source-descended transport-weighted imbalance quantity satisfies the active multiplicative window-to-window contraction condition on the NS safe tail: there exists a symbolic $\tilde{\kappa} \in (0, 1)$ — positivity supplied by the burden-(f) contraction domain, with the extinction case $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$ treated separately in the proof — (the route-conditional contraction ratio $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2)$; c_1 , c_2 , and the boundary coefficient η are certified, while η_{int} and η_R are supplied only by the hypothetical renewal-control discharge) such that*

$$\mathfrak{D}_{n+1} \leq \tilde{\kappa} \mathfrak{D}_n \quad \text{for all } n \text{ on the late safe tail.}$$

Hence the multiplicative-contraction clause of NS.6.1 (Theorem 6.3) holds conditionally.

PROOF. By Theorem 9.20 under its hypothetical renewal-control discharge (Obl. 9.5, part (e)), the total-renewal bound $\Psi_n \leq \tilde{\eta} \Phi_n$ gives the *Book II* ledger recurrence $\mathcal{D}_{n+1} \leq \mathcal{D}_n - (1 - \tilde{\eta})\Phi_n$. This is a $\mathcal{D}$ -statement; no $\mathfrak{D}$ -recurrence follows from it in current canon. The passage to the branch state is supplied *only* by the ledger-state map of burden (f) under the same hypothesised discharge: that map transfers the loss/renewal recurrence to $\mathfrak{D}_{n+1} \leq \mathfrak{D}_n - \Phi_n + \Psi_n$ and the H1/H2 lower bounds to the same state ($\Phi_n \geq (c_1 + c_2)\mathfrak{D}_n$), whence $\mathfrak{D}_{n+1} \leq \mathfrak{D}_n - (1 - \tilde{\eta})(c_1 + c_2)\mathfrak{D}_n = \tilde{\kappa} \mathfrak{D}_n$ with $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2)$. The certified bounds give $\tilde{\kappa} < 1$; the domain $\tilde{\kappa} > 0$ is part of the burden-(f) contraction domain (if instead $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$, then nonnegativity forces $\mathfrak{D} = 0$ in finitely many steps — the extinction case, in which the contraction condition of Theorem 6.3 holds trivially). In either case, we get $\mathfrak{D}_{n+1} \leq \tilde{\kappa} \mathfrak{D}_n$ with $\tilde{\kappa} \in (0, 1)$, or finite extinction. This is the multiplicative window-to-window contraction condition of Theorem 6.3. $\square$ $\square$

COROLLARY 12.2 ([C]). (NS.6.1 Reduction.) *Under the conditional SB2 discharge, NS.6.1 reduces to:*

- (1) Explicit $\tilde{\kappa}$ computation: *certify $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2)$ explicitly. The boundary/geometry quantities c_1, c_2, η are certified (Remark 9.21 and the quantitative propositions), but $\tilde{\eta} = \eta + \eta_{\text{int}} + \eta_R$ also requires the route coefficients $\eta_{\text{int}}, \eta_R$, supplied only by discharging the renewal-control route (Obl. 9.5, part (e)); until then $\tilde{\kappa}$ is symbolic.*
- (2) Promotion to target scope: *show the contraction is uniform across the full declared NS regime (not just the safe tail), which requires extending the SB2 argument from the safe-tail window to the global NS architecture.*

Item (1) becomes a finite quantitative computation only after the renewal-control route is discharged; its c_1, c_2, η part is complete. Item (2) is a remaining structural burden of NS.6.1, alongside the controlling renewal-control route itself.

REMARK 12.3 ([R]). **NS closure chain status (current).** The full NS discharge chain now stands, at conditional force, as:

$$\underbrace{\text{Ren.5 } [C]}_{\text{Ren.5a-e}} \Rightarrow \text{Ren.4 } [C] \Rightarrow \text{SB1e} \Rightarrow \text{SB1d} \Rightarrow \text{SB1c} \Rightarrow \text{SB1 } [C]$$

$$\text{SB2 } [C] \Rightarrow \text{NS.6.1 (partial) } [C] \Rightarrow \text{NS.6.1 (full) } [C] \Rightarrow \text{NS.7.1 } [C]$$

Every link in the chain is conditionally discharged: NS.6.1 at full scope (Cor. 13.9) and NS.7.1 (Thm. 7.1, Cor. 7.3), yielding the CERT-PROJ posture recorded in Prop. 15.1 (MIG-052; the former conditional CERT-CLOSE wording is superseded). The residuals are the external Stage-3 question (unconditional global regularity) and the internal common-state bridge (Obl. 9.5), through which every link's transfer to the window obstruction passes. *Transition snapshot (as recorded when this remark was first written; retained as intake — migration MIG-014):* at that stage NS.6.1 (full) and NS.7.1 were still open, and the remaining items were quantitative c_1, c_2, η certification, full-scope

promotion of NS.6.1, and NS.7.1; the quantitative certification is supplied immediately below.

PROPOSITION 12.4 ([C]). (Explicit c_1 Bound from Quotient Discipline.) [Conditional on Book I quotient discipline, LS-2, $K_0 = 7$.] *In the NS active architecture, the quotient-loss rate satisfies*

$$c_1 \geq \frac{1}{K_0} = \frac{1}{7}.$$

PROOF. By Book I quotient discipline, each non-quotient-visible boundary-mismatch datum is erased in exactly one transfer step (it fails to descend to the quotient class and is assigned to the null sector). The fraction of boundary-mismatch content that is non-quotient-visible on any single step is bounded below by $1/K_0 = 1/7$: since the collar alphabet has $K_0 = 7$ symbols, at most $K_0 - 1 = 6$ mismatch classes can be quotient-visible at any given stage; the remaining fraction $\geq 1/K_0$ is erased. This gives $c_1 \geq 1/7$. $\square$

PROPOSITION 12.5 ([C]). (Explicit c_2 Bound from Finite Collar Alphabet.) [Conditional on $K_0 = 7$, NS active window structure.] *In the NS safe-tail window with k active boundary sites, the finite-collar mixing rate satisfies*

$$c_2 \geq K_0^{-k}.$$

For the minimal NS window ($k = 2$): $c_2 \geq 1/49$.

PROOF. Two incompatible boundary-mismatch patterns on k active sites must collide (share a collar configuration) within K_0^k steps, by the finite-alphabet pigeonhole principle. At the collision step, the mixing contribution is at least $\mathfrak{D}_n/K_0^k$ (the full mismatch scale divided by the number of possible configurations). For the minimal NS window with $k = 2$ boundary sites: $c_2 \geq K_0^{-2} = 1/49$. More refined windows give larger c_2 . $\square$

PROPOSITION 12.6 ([C]). (Explicit η Bound from PASS Screening.) [Conditional on PASS derived (Thm. ??, GR branch; imported here), Phase-A.] *The PASS-screening ratio on the NS safe tail satisfies*

$$\eta \leq 1 - \frac{1}{K_0} = \frac{6}{7}.$$

Hence $\eta < 1$ is certified.

PROOF. By Theorem ?? (GR Branch, imported): the PASS condition holds conditionally, and backbone observables in W_A commute with the full symmetry group $G = \text{SU}(K_0)$. Fresh boundary renewal on the NS safe tail enters through the halo (the G -non-invariant sector). The fraction of renewal that is G -invariant (and thus could coherently drive the backbone) is bounded by $1 - 1/K_0 = 6/7$, because G -invariant content in K_0 symbols occupies at most $(K_0 - 1)/K_0$ of the total by the averaging argument over the G -action. Therefore the renewal coupling to the backbone is bounded by $\eta \leq 6/7 < 1$. $\square$

THEOREM 12.7 ([C]). (NS.6.1 Contraction Ratio — Symbolic, Route-Conditional; Reconditioned MIG-052.) *Under the quantitative bounds above and a hypothetical discharge of the renewal-control route (Obl. 9.5, part (e)) supplying the coefficients $\eta_{\text{int}}, \eta_R$ with $\tilde{\eta} = \eta + \eta_{\text{int}} + \eta_R < 1$, the NS.6.1 multiplicative contraction ratio is*

$$\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2) \in (0, 1),$$

symbolic until that route is discharged — $\tilde{\kappa} < 1$ from the certified bounds, $\tilde{\kappa} > 0$ from the burden-(f) contraction domain (the extinction case $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$, which forces $\mathfrak{D} = 0$ in finitely many steps, is treated separately wherever consumed). Current canon certifies only $c_1 \geq 1/7$, $c_2 \geq 1/49$, and the boundary coefficient $\eta \leq 6/7$; it certifies neither η_{int} nor η_R , so no explicit numerical total-ledger contraction bound is presently certified. Superseded (MIG-052; retained as historical intake): the former content of this theorem displayed $\kappa = 1 - (1 - \eta)(c_1 + c_2) \leq 1 - 8/343 \approx 0.9767$ and claimed completion of the quantitative certification for NS.6.1; that display treated the boundary-only η as a total-renewal coefficient and is withdrawn from active use.

PROOF. With $\tilde{\eta} < 1$ supplied by the hypothesised route discharge (Obl. 9.5, part (e)) and $c_1 + c_2 > 0$ from the certified quantitative propositions, $(1 - \tilde{\eta})(c_1 + c_2) > 0$, hence $\tilde{\kappa} < 1$; and $\tilde{\kappa} > 0$ holds by the burden-(f) contraction domain hypothesised with the same discharge ($0 < (1 - \tilde{\eta})(c_1 + c_2) < 1$), so $\tilde{\kappa} \in (0, 1)$. The certified lower bounds on c_1, c_2 alone prove only $\tilde{\kappa} < 1$, not positivity. No numerical value is asserted: the former substitution of $\eta \leq 6/7$ into a total ratio used the boundary coefficient beyond its proved scope. $\square$ $\square$

REMARK 12.8 ([R]). NS quantitative summary (reconditioned MIG-052). The three boundary/geometry quantities are certified with explicit bounds:

$$\begin{aligned} c_1 &\geq 1/7, \\ c_2 &\geq 1/49 \text{ (min window; improves with larger } k), \\ \eta &\leq 6/7 \text{ (boundary renewal channel only).} \end{aligned}$$

These are *subsidiary* quantitative requirements. *No explicit total contraction ratio is presently certified:* the total-renewal coefficient $\tilde{\eta} = \eta + \eta_{\text{int}} + \eta_R$ requires the route coefficients $\eta_{\text{int}}, \eta_R$, supplied only by the open renewal-control route (Obl. 9.5, part (e)) — the controlling open burden. The former display $\kappa \leq 1 - 8/343 \approx 0.977$ used the boundary-only η as a total coefficient and is superseded (MIG-052). The remaining burdens for NS.6.1 (full scope) are that controlling route together with the global promotion from the safe-tail window to the full NS regime: the window-by-window contraction must extend uniformly across all windows in the NS architecture, including uniformity of the route data, not just the minimal safe-tail case.

13. NS Window-Uniformity Verification

This section verifies the window-uniformity hypothesis required by Theorem 13.8: that the NS safe-tail window geometry is eventually periodic and the coefficient bounds c_1, c_2, η are uniform across all sufficiently late windows.

THEOREM 13.1 ([C]). (NS Window Finite-State and Isometry Classification — re-conditioned MIG-052.) [Conditional on $K_0 = 7$, Phase-A, LS-2, NS active architecture declaration.] (i) *The NS safe-tail window configurations take values in a finite state space: the number of distinct canonical collar-labelled window configurations is at most $K_0^{|\partial W|} < \infty$.* (ii) *Any two safe-tail windows with the same canonical collar-labelled configuration are isometric under the canonical collar-type labelling.* This theorem does not assert eventual periodicity: finiteness alone yields recurrence, not periodicity; eventual periodicity is derived in Cor. 13.6 from this finite-state result together with the deterministic transition of Thm. 13.5.

PROOF. Step 1 (Finite collar-type state space). By Book II LS-2: the stabilized collar type alphabet Σ_∂ is finite with $|\Sigma_\partial| = K_0 = 7$. The NS active window structure is a sequence of windows, each described by a configuration of collar types on a finite boundary. The number of distinct window configurations is at most $K_0^{|\partial W|} = 7^{|\partial W|} < \infty$, where $|\partial W|$ is the number of boundary sites per window. This proves (i). (Note, MIG-052: the former Step 2 of this proof invoked the Pigeonhole Principle to conclude eventual periodicity; pigeonhole on a finite set yields only recurrence of some value, not $w_{n+T} = w_n$ for all late n . That step is withdrawn; eventual periodicity is *not* concluded here.)

Step 2 (Isometry classification). Under Phase-A transitivity ($K_0 = 7$ collar walk visits all symbols), the collar-type equivalence relation on the boundary is canonical, so two windows with the same collar-type configuration are isometric under the canonical collar-type labelling. This proves (ii). $\square$ $\square$

COROLLARY 13.2 ([C]). (Uniform Coefficient Bounds Across Windows.) *Under the finite-state/isometry classification of Theorem 13.1 and the eventual periodicity of Cor. 13.6 (which supplies the periodicity consumed here; finiteness alone would not): the boundary/geometry coefficients $c_1 \geq 1/7$, $c_2 \geq 1/49$, and $\eta \leq 6/7$ hold uniformly across all sufficiently late windows (not just the minimal window).* This certifies those three quantities only: no total-renewal ratio $\tilde{\kappa}$ is inferred here, because uniformity of the remaining route data ($\eta_{\text{int}}, \eta_R$ of the hypothetical renewal-control discharge) is not supplied by window geometry. The geometric part of the window-uniformity hypothesis of Theorem 13.8 is verified; its route part remains hypothetical (MIG-052).

PROOF. By Cor. 13.6 (finite state space of Thm. 13.1(i) plus the deterministic transition of Thm. 13.5): the window sequence is eventually periodic, so all sufficiently late windows belong to the finitely many isometry classes along the cycle (Thm. 13.1(ii)). The coefficient bounds c_1, c_2, η depend only on K_0 and the window boundary geometry (Propositions 12.4–12.6), which is uniform across isometric windows. Hence the bounds are uniform. $\square$ $\square$

REMARK 13.3 ([R]). NS.6.1 geometric uniformity condition closed; route condition open (MIG-052). The *geometric* window-uniformity hypothesis of Theorem 13.8 is verified by Theorem 13.1 and Corollary 13.2. The NS.6.1 global contraction theorem is conditional on Phase-A, $K_0 = 7$, LS-2, the NS architecture declaration, *and* the hypothetical renewal-control discharge with uniform route data (Obl. 9.5, part (e));

no additional external *geometric* uniformity hypothesis is required. The NS branch closure chain is:

$$\text{Ren.5 } [C] \Rightarrow \dots \Rightarrow \text{SB2 } [C] \Rightarrow \text{NS.6.1 } [C] \Rightarrow \text{NS.7.1 } [C] \Rightarrow \text{CERT-PROJ } [C].$$

All steps now carry the same conditional hypotheses: Phase-A, $K_0 = 7$, LS-2, PASS (derived), and the NS architecture declaration — and, per MIG-052, the terminal step holds at CERT-PROJ force only: the ledger-to-obstruction transfer is the open common-state bridge (Obl. 9.5).

REMARK 13.4 ([R]). Micro-hardening note: Deterministic Finite Transition Lemma. The pigeonhole argument in Thm. 13.1 shows that the safe-tail window configuration sequence takes values in a finite set and therefore must repeat some state. For the conclusion of *eventual periodicity* — not merely recurrence — the proof requires that window transitions be deterministic. This was the named hardening item (Q2a). The MSC uniqueness argument supplies it: if the NS active architecture declaration implies a unique minimal faithful support for each window-to-window transition, the transition is deterministic.

THEOREM 13.5 ([C]). (NS Deterministic Window-Transition.) [Conditional on: the NS active architecture declaration; the canonical collar-type labelling of window configurations; and the declared finite transport/defect update rule.] *The next safe-tail window configuration is uniquely determined by the current canonical collar-labelled window configuration together with the declared finite transport/defect update rule. That is, the window-configuration transition function is a well-defined map*

$$\sigma : \mathcal{W}_{\text{safe}} \rightarrow \mathcal{W}_{\text{safe}}$$

on the finite window-configuration state space.

PROOF. Define the *window-transition carrier* as the minimal source-visible support needed to determine the next window configuration from the current one and the Book III update rule. By the NS active architecture declaration, every admitted window configuration carries a canonical collar-type label, and the transport/defect update rule is declared with no hidden channel (Book III, no-fourth-channel discipline).

Suppose two NS active architecture declarations yielding the same current window configuration W produced distinct next window configurations $W'_1 \neq W'_2$. Then some admissible probe witnesses a difference in the resulting window data. By the exhaustive source-image condition (**def:exhaustive-source-image**, Book IV) applied to $\mathbf{U}$, this difference would require an undeclared window-transition channel beyond the declared collar-type label and transport/defect update rule. No such channel is declared. Therefore $W'_1 = W'_2$, and the transition is unique. $\square$ $\square$

COROLLARY 13.6 ([C]). (Eventual Periodicity of Window Configurations.) [Conditional on Thm. 13.5 and Thm. 13.1.] *The safe-tail window configuration sequence is eventually periodic: finite state space plus deterministic transition implies eventual cycle.*

PROOF. By Thm. 13.1(i), the sequence takes values in a finite set (finiteness alone would yield only recurrence). By Thm. 13.5, the transition function σ is well-defined

and deterministic. A deterministic map on a finite set must eventually enter a cycle: once any configuration repeats, determinism forces the entire subsequent segment to repeat. Hence the sequence is eventually periodic, and the isometry classification of Thm. 13.1(ii) applies along the cycle. $\square$ $\square$

REMARK 13.7 ([R]). Q2a resolved. Thm. 13.5 and Cor. 13.6 close the Q2a micro-hardening gap. Thm. 13.1 now carries clean [C] status in its reconditioned finite-state/isometry form: finite state space, deterministic transition (Thm. 13.5), eventual cycle (Cor. 13.6), uniform boundary/geometry coefficient bounds (Cor. 13.2). The global NS.6.1 contraction remains *route-conditional*: it consumes the hypothetical common-state discharge (Obl. 9.5) and stands at CERT-PROJ force while that burden is open (MIG-052).

Section 12 established the *route-conditional, symbolic* contraction $\tilde{\kappa} \in (0, 1)$ on the minimal safe-tail window — conditional on the hypothetical common-state discharge (Obl. 9.5); the certified quantities are the boundary/geometry bounds c_1, c_2, η only, and no numerical total ratio is certified (MIG-052). This section promotes that conditional contraction to the full NS regime on the same hypothetical footing.

THEOREM 13.8 ([C]). (NS.6.1 Full Scope: Uniform Window-to-Window Contraction.) [Conditional on the symbolic route-conditional ratio of Theorem 12.7 — hence on the hypothetical renewal-control discharge (Obl. 9.5, part (e)) with its route data $\eta_{\text{int}}, \eta_R$ uniform across the declared windows — and the NS window-uniformity hypothesis: the window geometry and the safe-tail active architecture are uniform across all sufficiently late windows. Uniformity of c_1, c_2, η alone is insufficient.] *The multiplicative contraction $\mathfrak{D}_{n+1} \leq \tilde{\kappa} \mathfrak{D}_n$ holds uniformly across all windows in the NS active architecture for all sufficiently large n , not just within the minimal safe-tail window.*

PROOF. Three steps:

Step 1 (Uniform window geometry). By Cor. 13.6 (the finite configuration-state space of Thm. 13.1(i) together with the deterministic transition of Thm. 13.5), the active window structure on the NS safe tail is eventually periodic: for all sufficiently large n , the n -th window is isometric to the $(n + T)$ -th for some fixed period $T \geq 1$ (Thm. 13.1(ii)). This gives uniform geometry across windows.

Step 2 (Uniform coefficient bounds). Under uniform window geometry:

- $c_1 \geq 1/7$ holds uniformly (Proposition 12.4): the quotient-loss rate depends only on K_0 and Book I quotient discipline, both of which are uniform across windows.
- $c_2 \geq 1/49$ holds uniformly (Proposition 12.5): the mixing rate depends only on K_0 and the window size k , both uniform.
- $\eta \leq 6/7$ holds uniformly (Proposition 12.6): the PASS screening ratio depends only on K_0 and the PASS condition, which is uniform by Theorem ?? (imported from GR branch).

Step 3 (Global contraction). The uniform boundary/geometry bounds, together with the hypothesised uniform route data $\eta_{\text{int}}, \eta_R$ supplied by the renewal-control discharge (Obl. 9.5, part (e)), give the same symbolic contraction ratio $\tilde{\kappa} =$

$1 - (1 - \tilde{\eta})(c_1 + c_2) \in (0, 1)$ at every sufficiently late window ($\tilde{\eta} = \eta + \eta_{\text{int}} + \eta_R < 1$; positivity from the burden-(f) contraction domain; no numerical value is certified while the route is open; in the extinction case $(1 - \tilde{\eta})(c_1 + c_2) \geq 1$ nonnegativity forces $\mathfrak{D} = 0$ in finitely many steps and the conclusion holds trivially). In the geometric case, by induction, $\mathfrak{D}_n \leq \tilde{\kappa}^{n-n_0} \mathfrak{D}_{n_0} \rightarrow 0$ as $n \rightarrow \infty$. Hence the contraction holds globally, at CERT-PROJ force while the route is open. $\square$ $\square$

COROLLARY 13.9 ([C]). (NS.6.1 Conditionally Discharged.) *Theorem 6.3 (the Obstruction-to-Continuation-Criterion Theorem NS.6.1) is conditionally discharged at CERT-PROJ force (MIG-052): the multiplicative window-to-window contraction condition holds uniformly on the ledger side; its transfer to the window obstruction, and hence the certification of the active NS persistence/extendibility continuation criterion, passes through the open common-state bridge (Obl. 9.5).*

The conditional hypotheses are:

- (1) *NS safe-tail window-uniformity (eventually periodic window geometry);*
- (2) *$K_0 = 7$ (unconditional by Thm. ??);*
- (3) *Phase-A (conditional on $K_0 = 7$);*
- (4) *PASS (conditional theorem, Thm. ??);*
- (5) *IDC — Interior Defect Control (1)–(2) (subsidiary and insufficient per the MIG-052 re-scope; MIG-050 intake);*
- (6) *the obstruction/ledger common-state bridge, Obl. 9.5 (open obligation; MIG-052). The former hypothesis at this position — α , obstruction/ledger alignment (**hyp:NS-alpha**) (MIG-050) — is withdrawn as ill-typed (MIG-052).*

PROOF. Theorems 13.8 and 9.20 (reconditioned) supply the ledger contraction and the SCT.6b separation; their transfer to the window obstruction, and the satisfaction of the contraction condition itself (Thm. 9.4), are conditional on hypotheses (5)–(6) — in particular on the open common-state bridge Obl. 9.5 — and the ledger ratio is symbolic, route-conditional (Thm. 12.7, reconditioned). The continuation criterion of Theorem 6.3 follows on the stated hypothesis set. $\square$ $\square$

REMARK 13.10 ([R]). **NS.7.1: from final open frontier to conditional discharge (transition snapshot).** *As recorded at the time this remark was first written (retained as a dated transition snapshot — migration MIG-014):* with NS.6.1 conditionally discharged, NS.7.1 (the branch convergence/endpoint theorem) was the only remaining open obligation before domain-bounded CERT-CLOSE — a structural theorem about the NS endpoint functional, not a new quantitative obligation. *Current status:* that step has since been taken. NS.7.1 is conditionally proved (Thm. 7.1, Cor. 7.3), and the closure chain is complete at conditional force:

$$\text{Ren.5 } [C] \Rightarrow \cdots \Rightarrow \text{SB2 } [C] \Rightarrow \text{NS.6.1 } [C] \Rightarrow \text{NS.7.1 } [C],$$

yielding the CERT-PROJ posture of Prop. 15.1 (MIG-052; the former conditional CERT-CLOSE wording is superseded) — the hypothesis set now comprises the open common-state bridge obligation (Obl. 9.5) with IDC subsidiary; the former hypothesis α is withdrawn (see Thm. 9.6). The residuals are the external unconditional Stage-3 question and the internal common-state bridge.

14. H-LR: Lieb–Robinson Bound and NS Stage-3 Canonical Import

THEOREM 14.1 ([C]). (H-LR: Lieb–Robinson Bound for the Fluid Sector.) [Conditional on $K_0 = 7$, Phase-A, O-CLU discharged (YM Branch, Thm. ??).] *The fluid sub-algebra $A_{\text{fluid}} \subseteq A_\infty$ of the NS observable algebra is identified with the screened sector W_A . The Lieb–Robinson bound holds on W_A : for observables $\mathcal{O}_1, \mathcal{O}_2$ supported in regions separated by distance d ,*

$$|[\mathcal{O}_1(t), \mathcal{O}_2]| \leq C \|\mathcal{O}_1\| \|\mathcal{O}_2\| e^{-\mu(d - v_{\text{LR}}|t|)},$$

with $\mu = m^2\beta > 0$ (from O-CLU exponential clustering, $\beta = \log_2(3/2)$, $m^2 > 0$ from the YM mass gap) and Lieb–Robinson velocity v_{LR} .

PROOF. Identification $A_{\text{fluid}} = W_A$. The fluid observable sub-algebra is the sub-algebra of A_∞ consisting of observables that (i) are G -invariant (PASS condition: $[a, U_g] = 0$ for all $g \in G$, Thm. ??, GR Branch) and (ii) are screened-sector-localizable (supported on W_A). Both conditions characterize exactly the screened sector $W_A = \ker(T_\partial - \lambda_{\max} I)^\perp$ (Book V, Thm. ??). Hence $A_{\text{fluid}} = W_A$.

Lieb–Robinson from exponential clustering. By O-CLU (Thm. ??, YM Branch, imported): connected correlators decay exponentially on W_A : $|\langle \mathcal{O}_1 \mathcal{O}_2 \rangle_c| \leq C e^{-\mu d}$ with $\mu = m^2\beta > 0$. The Lieb–Robinson bound follows from exponential clustering by the standard Lieb–Robinson argument (Lieb–Robinson 1972): the commutator $[\mathcal{O}_1(t), \mathcal{O}_2]$ is bounded by the exponentially decaying connected correlator at spatial separation $d - v_{\text{LR}}|t|$. The LR velocity is $v_{\text{LR}} = 2e/\mu$ (Hastings–Koma bound, classical). $\square$

THEOREM 14.2 ([C]). (H-CODIM2: LR-Weighted Depth-Sum Bound.) [Conditional on Thm. 14.1.] *Under the Lieb–Robinson bound on W_A , the LR-weighted depth-sum satisfies the boundary-order estimate:*

$$\sum_{d \geq 0} |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|),$$

where $\partial_d U_n$ is the set of sites at distance d from ∂U_n . Hence the NS certified-event count satisfies the boundary-order (Strong BA) bound: $N_{\text{cert}}(U_n, t) \leq C_\partial |\partial U_n|$.

PROOF. The LR weight $e^{-\mu d}$ makes the depth-sum convergent; the boundary area $|\partial_d U_n|$ grows at most polynomially in d (by the isoperimetric inequality on the collar substrate). Hence $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$ by geometric series comparison. Strong BA follows from this depth-sum bound and the definition of the certified-event count (the events with LR-weighted contribution to the depth-sum). $\square$

COROLLARY 14.3 ([C]). (NS Stage-3 Conditional Discharge.) [Conditional on Theorems 14.1 and 14.2.] *Strong BA ($N_{\text{cert}} = O(|\partial U_n|)$) implies:*

$$\sup_n Q_n := \sup_n D_{\text{floor}}(U_n)/a_n^2 < \infty,$$

closing NS Stage-3. Combined with the canonical NS closure chain (NS.6.1 [C] $\rightarrow$ NS.7.1 [C]; per MIG-052 that chain holds at CERT-PROJ force, its obstruction transfer passing through the open common-state bridge, Obl. 9.5), the NS branch achieves $u \in$

$C^\infty(\mathbb{R}^3 \times (0, \infty))$ conditionally at status $\text{Cond}(K_0 = 7)$, on the same enlarged, bridge-inclusive hypothesis set.

PROOF. By Thm. 14.2: $N_{\text{cert}} = O(|\partial U_n|)$. Under Van Hove exhaustion, $a_n^2 = (2\pi/K_0^n)^2 \rightarrow 0$ and $|\partial U_n| = O(K_0^{2n/3})$ (surface-to-volume scaling). The quotient $D_{\text{floor}}/a_n^2 = N_{\text{cert}}/T_{\text{cov}} \cdot a_n^{-2}$ is bounded because the LR localization forces the certified-event density to be boundary-dominated rather than volume-dominated, exactly matching the a_n^2 decay. $\square$ $\square$

REMARK 14.4 ([R]). **NS canonical status after H-LR import.** The import of H-LR (Thm. 14.1) and H-CODIM2 (Thm. 14.2) into the canonical NS Branch completes the H-ladder inputs of the conditional Stage-3 chain. *This does not close NS Stage-3, even conditionally, on its own (MIG-052):* the chain's transfer to the window obstruction — hence any Stage-3 reading — passes through the controlling open common-state burden (Obl. 9.5), and the branch remains at CERT-PROJ force with that frontier open. The conditional chain is now:

$$A_{\text{fluid}} = W_A [C] \Rightarrow \text{LR bound } [C] \Rightarrow \text{Strong BA } [C] \Rightarrow \sup_n Q_n < \infty [C] \Rightarrow \text{CERT-PROJ } [C]$$

All steps conditional on Phase-A, $K_0 = 7$, O-CLU, and the YM mass gap — and, per MIG-052, on the open common-state bridge (Obl. 9.5), through which the terminal composition with NS.6.1/NS.7.1 passes; the terminal token is accordingly CERT-PROJ, not CERT-CLOSE. This records the 8-step pre-canonical resolution chain (Papers BW + BX + BS) in canonical form at that force.

15. Final Status Proposition

PROPOSITION 15.1 ([C]). (NS Branch Status Proposition — Updated, MIG-052.) *The NS branch is a lawful descendant of the canonical spine and holds a domain-bounded conditional endpoint claim at CERT-PROJ force: the NS.6.1/NS.7.1 conditional chain is complete on the ledger side, and its transfer to the window obstruction — hence the endpoint — passes through the open common-state bridge obligation (Obl. 9.5). The conditional hypothesis set comprises — in addition to Phase-A, $K_0 = 7$, LS-2, and PASS — the bridge obligation together with the subsidiary, insufficient IDC (1)–(2); the former hypothesis α (**hyp:NS-alpha**) (MIG-050) is withdrawn. The former domain-bounded conditional CERT-CLOSE wording (MIG-050) is superseded; no claim in this proposition may be cited at CERT-CLOSE force, nor without the enlarged, bridge-inclusive hypothesis set.*

PROOF. Lawful descent: Proposition 3.1. NS.6.1 conditionally discharged: Cor. 13.9 (window-uniformity + $K_0 = 7$ + Phase-A + PASS + IDC (subsidiary) + Obl. 9.5; the ledger contraction stands in its symbolic route-conditional form $\tilde{\kappa} = 1 - (1 - \tilde{\eta})(c_1 + c_2) < 1$ — the former explicit $\kappa \leq 1 - 8/343$ used the boundary-only η as a total coefficient and is superseded (MIG-052) — its transfer to the window obstruction being exactly the open common-state bridge; α withdrawn; the former per-step margin claim remains withdrawn). Window determinism: Thm. 13.5 (Q2a resolved), giving clean [C] without recurrence caveat. NS.7.1 conditionally discharged: Thm. 7.1 (Book VI H_x^1 interface

+ Grönwall control). Domain-bounded conditional endpoint at CERT-PROJ force: Cor. 7.3. The NS Frontier Theorem (Thm. 8.2) now describes the *formal conditional* frontier, not an unconditional barrier: under the stated Phase-A/ $K_0 = 7$ /PASS/LS-2 hypotheses *together with the hypothetical discharge of the controlling common-state burden* (Obl. 9.5: *total-renewal control, same-state obstruction comparison, ledger-state map with positive contraction domain*), the branch would reach its declared conditional endpoint; absent that discharge it stands at CERT-PROJ force with the common-state frontier open (MIG-052). Unconditional global regularity (NS Stage-3 Clay Prize) remains the external frontier. $\square$ $\square$

COROLLARY 15.2 ([U]). *The NS branch may claim:*

- (1) *lawful source descent from $\mathbf{U}$,*
- (2) *lawful branch admissibility under Book V,*
- (3) *a well-defined source-descended obstruction quantity,*
- (4) *a lawful bridge-stack formulation whose named theorems (NS.6.1, NS.7.1) are conditionally discharged at CERT-PROJ force (Cor. 13.9, Thm. 7.1, Cor. 7.3) — conditionally on the enlarged hypothesis set including the open common-state bridge obligation Obl. 9.5 with IDC subsidiary (MIG-052; α withdrawn),*
- (5) *a domain-bounded conditional endpoint claim at CERT-PROJ force at the declared scope (Prop. 15.1), conditional on the same enlarged, bridge-inclusive hypothesis set (MIG-052; the former conditional CERT-CLOSE wording is superseded), and*
- (6) *an explicit named failure frontier: originally Ren.5 and SB2, both now conditionally discharged; the residual frontier comprises the external unconditional Stage-3 question (unconditional global regularity) and the internal frontier as sharpened by the MIG-052 common-state analysis: the open common-state bridge obligation Obl. 9.5, with IDC (1)–(2) recorded under it as subsidiary and insufficient, and the withdrawn hypothesis α (**hyp:NS- α**) retained only as historical intake.*

The NS branch may not claim unconditional endpoint closure, unconditional global regularity, or CERT-CLOSE at any scope while Obl. 9.5 is open.

REMARK 15.3 ([R]). (Pre-canonical structural achievements.) The v8.41 pre-canonical monograph recorded derivations of the following results. These derivations have no canonical theorem-bearing force; canonical force requires re-derivation in canonical papers. For the record:

- (a) **NS Stage-0** (Thm. 36.3): thermodynamic floor density $f_\infty > 0$ exists unconditionally under N-add, T-add, BV, and non-degeneracy.
- (b) **NS Stage-1** (Thm. 36.4): boundary flux $\varphi_\infty \geq 0$ stabilizes unconditionally.
- (c) **NS Stage-2** (Thm. 36.6): discrete balance identity and nonlinear term identification $(u \cdot \nabla)u + \nabla p$ from Weyl product rule (conditional on KPO-3).
- (d) **Leray weak existence** (Thm. 36.13): for every $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, a global Leray weak solution exists unconditionally.

- (e) **NS Stage-3 pre-canonical path:** under $K_0 = 7$ (a theorem, Bk. I Thm. ??), the dream-suite discharge chain (Papers BW + BX + BS) gives $u \in C^\infty(\mathbb{R}^3 \times (0, \infty))$ at status $\text{Cond}(K_0 = 7)$. See Remark 8.4.

Items (a)–(d) have recorded pre-canonical derivations at stated-unconditional level; canonical force requires re-derivation in canonical papers. Item (e) records a conditional closure path pending Theorem 6.3; the path has no canonical force.

REMARK 15.4 ([R]). (Current canonical re-derivation state.) The following results have now been re-derived in canonical papers:

- NS Stage-0: $f_\infty > 0$ exists (Thm. 4.9, [U]).
- NS Stage-1: $\varphi_\infty \geq 0$ exists (Thm. 4.10, [U]).
- NS Stage-2a: Discrete balance identity (Thm. 4.11, [U]).
- NS Stage-2b: Nonlinear term identification (Thm. 4.12, [C] on KPO-3).
- Leray weak existence: global L^2 weak solution (Thm. 4.13, [U]).

NS Stage-3 (Thm. 6.3) remains [O] in canonical papers.

16. Boundary of the NS Branch Book

What the NS branch book establishes.

- (1) The NS branch is a lawful descendant of **U**, satisfying all five conditions of the Universal Branch Legitimacy Theorem.
- (2) The NS obstruction quantity is well-defined and source-descended, not an external primitive.
- (3) The bridge stack from source-descended obstruction control to the NS endpoint target is completely specified, with the missing step named precisely as NS.6.1.
- (4) The proof program for NS.6.1 is structured and explicit, with the current frontier narrowed to UH.3 on the spike-free proof-program side and SB2 on the reserve-side structural side.
- (5) Strong Boundary Safety is not the continuation theorem in disguise; it is strictly upstream of it.

What the NS branch book does not establish.

- (1) *NS closure.* The NS Frontier Theorem is unconditionally proved; NS.6.1 is not.
- (2) *Proof of the full conditional package.* The branch still lacks discharge of the last proof-program micro-lemma UH.3 or explicit adoption of its spike-free-side dissipation replacement, and it still lacks unconditional reserve on the safe tail beyond SB2.
- (3) *Resolution of the NS Millennium Prize question.* That reduction is downstream of both UH.3/SF and SB2, together with the endpoint bridge NS.7.1.
- (4) *Any source-level claim.* No theorem in this branch book modifies or supplements the content of Books I–VII.

REMARK 16.1 ([R]). For the branch's operative conditional theorem package, see Section 10. For the final residual frontier after all reductions, see the current frontier note in Section 9.6 and the status statement in Section 12.

17. Dependency Ledger

Theorem	St.	Depends on	Exports to
3.1 NS Lawful Descent	[U]	Bk. IV, Bk. V	3.2, 8.2
3.2 NS Legitimacy Witness	[U]	Bk. V	3.3, 8.2
4.7 Source-to-NS Obstr.	[U]	Bk. III, Defs. 4.3–4.6	6.3, 8.2
5.3 Criterion Required	[U]	Bk. VI	5.4, 6.3
9.1 Window Coherence	[U]	Bk. III, Def. 6.1	9.2
9.2 Successor De-comp.	[C]	9.1, Bk. III three-source	9.4
9.8 Internal Defect Subord.	[C]	9.2, Phase-A floor	9.6
9.12 Late Align.-to-Dissip.	[C]	Active architecture	SB1 chain $\Rightarrow$ 9.10
9.14 Positive Reserve	[C]	Spike-free architecture	9.22
9.10 Spike-Res. Bdry Subord.	[C]	9.12, 9.14	9.6
9.6 Split-Defect Subord.	[C]	9.8, 9.10	9.4
9.22 Strict Gap Est.	[C]	9.6, transport ceiling	9.4
9.4 Above-Threshold Contr.	[C]	9.2, 9.6	6.3
6.3 Obstr.-to-Criterion Thm.	[C]	9.1–9.4, Def. 5.2	6.5, 7.3
7.1 Criterion-to-Endpoint	[C]	Def. 5.2, Bk. VI	7.3
8.2 NS Frontier Thm.	[U]	3.1, 5.3, 6.3, Bk. VI, Bk. VII	8.3, 8.5
15.1 NS Status Prop.	[C]	3.1, 8.2	Branch summary; external status statement

SCHOLIUM 17.1 ([R]). The NS branch is lawful, structurally serious, and precisely organized. It is not blocked by a vague missing bridge. The frontier items that once blocked it — UH.3 on the proof-program side and SB2 on the reserve-side structural side, each with a precise mathematical statement, a clear logical role in the proof graph, and an explicit anti-circularity argument — have been conditionally discharged, and the branch holds its domain-bounded conditional endpoint claim at CERT-PROJ force (Prop. 15.1; MIG-052). What remains is a pair of named residuals: the *internal* obstruction/ledger common-state bridge (Obl. 9.5), through which every transfer from the contracting ledger to the window obstruction passes, and the *external unconditional* frontier — unconditional global regularity (the Stage-3 Clay question) — which lies outside the declared conditional scope and is claimed nowhere in this book. That precision — named items, named conditions, named residual — is the discipline Book VII mandates, and the NS branch book satisfies it.

REMARK 17.2 ([R]). (Note on the pre-canonical record.) This note concerns the *pre-canonical* v8.41 record and its relation to the canonical body; the closing assessment of the canonical branch itself is Scholium 17.1 above. The pre-canonical program has recorded derivation sketches for every structural sub-obligation except NS Stage-3 (the Clay Prize question). These derivation sketches have no canonical force; they identify which canonical proofs remain to be written. All obligations within reach of the current NFC framework are discharged. The frontier Theorem 6.3 is not vague: it asks whether the LR-weighted boundary-order condition (Definition 2.6) holds, and the dream-suite resolution path shows it does under $K_0 = 7$. That import is now reflected in the canonical body: NS.6.1 is conditionally discharged (Cor. 13.9), NS.7.1 is conditionally proved (Thm. 7.1, Cor. 7.3), and the branch holds its domain-bounded conditional endpoint claim at CERT-PROJ force (Prop. 15.1; MIG-052 — the transfer to the window obstruction passes through the open common-state bridge, Obl. 9.5). Unconditional global regularity (NS Stage-3, the Clay Prize question) and the internal common-state bridge are the named residual frontiers.